Jashore University of Science and Technology

Jashore-7408, Bangladesh

EV Charging Load Demand Prediction Using Machine Learning in a SUMO-Based Simulation Environment

Submitted By:

Rakesh Biswas

Student ID: 200112

Registration Year: 2022

Academic Year: 2020–2021

Supervisor:

Monishanker Halder

Assistant Professor, Department of CSE

Jashore University of Science and Technology

This thesis report was submitted in partial fulfillment of the

Requirements for the Bachelor's Degree

In **Computer Science and Engineering**

Faculty of Computer Science and Engineering

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27 April 2026

Jashore University of Science and Technology

Jashore-7408, Bangladesh

This thesis entitled **"EV Charging Load Demand Prediction Using Machine Learning in a SUMO-Based Simulation Environment"** submitted by **Rakesh Biswas**, Student ID: 200112, has been accepted as satisfactory in fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering.

Signature of the Student

Rakesh Biswas

Signature of the Supervisor

Dedication

To my beloved parents, whose endless love, sacrifice, and encouragement
have been the cornerstone of every achievement in my life.

To my teachers, who ignited the flame of curiosity within me
and guided me towards the pursuit of knowledge.

And to all the researchers and engineers working tirelessly
towards a sustainable, electrified future of transportation.

Acknowledgments

At First, I want to express my heartfelt gratitude and sincere thanks- to my supervisor, **Monishanker Halder**. And this thank is just not for his invaluable guidance but also for his constant encouragement and expert advice accross this research-work. His constructive feedback and unwavering support played a key role in shaping this work.

I would like to express my sincere gratitude- to the faculty members of the Computer Science and Engineering (CSE) Department of Jessore University of Science and Technology (JUST) for giving an intellectually stimulating environment and their continuous support during my graduate studies.

In the last, I am immensely grateful- to my family for their patience, prayers, and moral support. Without them, this work would not have been possible.

Rakesh Biswas

Department of CSE, JUST

27 April 2026

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Abstract

The rapid growth of Electric Vehicle (EV)s presents both an important opportunity and considerable challenge for modern power grid management. To optimize energy distribution, accurate prediction of EV charging load demand is critical. It reduces peak-hour stress on electrical infrastructure and enables smart grid operations. This thesis presents a comprehensive framework for predicting EV charging load demand using a suite of supervised Machine Learning (ML) and Deep Learning (DL) models applied to data generated from a realistic microscopic traffic simulation. To enable real-time vehicle monitoring and intelligent charging station management, the simulation environment was built by integrating the open-source urban mobility simulator SUMO with the Traffic Control Interface (TraCI). A network of 10 bidirectional charging stations was deployed with configurable power ratings ranging from 25 kW to 50 kW. Also, separate charging and waiting queue slots were configured for each station. Forty-nine electric vehicles of the battery-powered type were simulated for 600 seconds. This resulted in 34,020 vehicle-level records, including features such as state of charge (SOC), speed, acceleration, station utilization, queue length, and system-wide load metrics. Seven different models were evaluated for system-level load demand prediction: Linear Regression, Decision Tree, Random Forest, XGBoost, Artificial Neural Network (ANN), Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU). The performances were measured using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the Coefficient of Determination (R^2). The results demonstrate that the ANN model achieved the highest precision with an R^2 score of 0.9967, followed closely by Linear Regression and tree-based ensemble methods. In addition, a rule-based smart charging recommendation system was implemented. In this scenario, vehicles were routed to the optimal charging station based on real-time SOC level, distance to the nearest station, slot availability and queue status. The findings show that SUMO-based simulation is an effective method for generating enriched EV charging datasets, and that high-capacity models like ANN are exceptionally effective for precisely modeling the complex nature of system-wide EV load demand.

Keywords: Electric Vehicle, Charging Load Prediction, Machine Learning, SUMO Simulation, Artificial Neural Network (ANN), Random Forest, XGBoost, Smart Charging, SOC, Queue Management.

Chapter 1

Introduction

Chapter 1: Introduction

The global transportation sector is undergoing a radical change due to the urgent need to reduce greenhouse gas emissions and to reduce dependence on fossil fuels. For totally decarbonizing road transport, Electric Vehicle (EV)s have emerged as one of the most promising technologies. According to the [1], the number of electric vehicles worldwide exceeded 26 million in 2022 and is forecast to continue to grow rapidly in the coming decades. With the rapid growth of electric vehicle usage, the increasing demand for electricity for charging is becoming a serious concern for both power system managers and urban planners [2]. Recently, advanced integrated frameworks and deep learning hybrids have been proposed to address the increasingly complex nature of these load patterns [3, 4].

With the rapid growth of EVs, the need for intelligent control infrastructure for energy stability and efficiency of optimal use has become significant. Three-wheeled electric vehicles such as Easy Bikes, Electric Rickshaws, and Electric Vans in developing countries like Bangladesh have accelerated the need for scalable charging solutions [5]. These vehicles play a major role in urban and semi-urban mobility. However, their unbalanced charging behavior can lead to sudden fluctuations in electricity demand, which creates pressure on the power grid.

EVs communicate directly with the power grid. Uncoordinated charging behavior—especially during peak hours—can create significant stress on the distribution network. This can lead to voltage violations, transformer overloads, and increased energy costs [6]. Therefore, effective management of EV charging loads requires accurate forecasting tools capable of estimating spatiotemporal demand patterns across the charging station network [7].

In this field, one of the key challenges is to accurately predict system-level charging load demand. Reliable load prediction enables better planning of charging stations, improved power distribution, and the prevention of overload situations.

1.1 Background

The rapid increasing of electric vehicles is reshaping transportation and electricity distribution systems. Unlike conventional fueled vehicles, electric vehicles, shift energy demand from fuel stations to the power grid . This creates loads that vary over time and space. Understanding the movement, energy consumption and charging behavior of electric vehicles is essential for load prediction, grid reliability and optimal utilization of charging infrastructure [7]. In cities of Bangladesh , a major drawback is the unorganized and untested charging behavior of electric vehicles . Vehicles oftenly crowd around specific nodes or roadside charging points. Which is resulting in large amounts of energy consumption during peak hours . This often results in stress on the local grid, voltage drops and overloading of transformers problems. Which that already exist in rapidly growing cities like Khulna, Jashore, Satkhira. Besides, energy consumption, charging-load varies greatly depending on vehicle speed, congestion levels, route selection, and driver behavior . Without proper modeling and prediction of these parameters, city authorities and electricity distributors face increasing difficulties in ensuring grid reliability, planning future charging stations and controlling unbalanced load distribution. This research work uses traffic simulation tools such as SUMO to model real-world urban mobility. Where EV extensions to SUMO help model battery behavior, charging events, energy consumption, system load behavior. These outputs are integrated with load prediction models to create a robust approach for analyzing the load impact of EV adoption.

1.1.1 Definition

1.1.1.1 Battery-Powered Vehicle (BPV) or EV

A Battery Powered Vehicle(BPV) or Electric Vehicle (EV) is a transportation system which is powered entirely by rechargeable batteries and electric motors instead of an internal combustion engine. EVs store electrical energy in lithium-ion or lead-acid batteries and through an electric drivetrain, they convert this stored energy into mechanical energy [2]. In the context of Bangladesh in urban areas like Khulna and Jessore -electric vehicles include the widely used low-cost electric three-wheelers (e.g., easy bikes , e-rickshaws, e-vans). Those vehicles are popular due to their low operating costs, absence of exhaust emissions and suitability for short-distance travel within cities or urban area . Their electricity-based operations directly link transportation demand to local electrical grid-loads. Which is making EV behavior a critical element in grid planning.

1.1.1.2 Load Prediction

Load prediction refers to the process of estimating the total electrical demand of a system based on historical or observed data. In this study, it specifically refers to the prediction of the system-level charging load (`system_total_load_kW`) generated by electric vehicles using machine learning models [8]. The prediction is performed by understanding the relationship between various input characteristics —such as time progress , number of vehicles charging, system-level vehicle statistics and battery charge status and the corresponding load demand . Unlike traditional analytical approaches , this research uses data-driven models to understand the complex and nonlinear dependencies within the system. Proper load prediction can help to understand in better way how electric vehicle charging behavior affects overall energy demand and helps in efficient planning and management of charging infrastructure.

1.1.1.3 Traffic Microsimulation

Traffic microsimulation is a precise, vehicle-level modeling technique that which captures the detailed movements of individual vehicles over time. It represents the interaction of speed, acceleration, deceleration, lane change behavior, path selection, stop-and-go cycles and congestion with high temporal accuracy. Microsimulation becomes increasingly important in EV research. As vehicle speed directly affects the dynamics of energy consumption and battery degradation. By integrating microsimulation with EV models , researchers can generate realistic estimates of power consumption, load behavior under different road, time and traffic conditions. Which ultimately helps in more accurate charging load analysis and prediction .

1.1.1.4 State of Charge (SOC)

State of Charge (SOC) is a quantitative measurement which indicates the remaining energy stored in an electric vehicle (EV) battery [9]. It is usually expressed as a percentage (0–100%). SOC decreases when the vehicle is being driven due to the use of propulsion power and it increases when the vehicle is connected to a charging station. As it determines when and where a vehicle will need charging , SOC is a fundamental variable for understanding electric vehicle behavior . In simulation environments like SUMO, SOC-logs give : time-step-level insights into real-time power consumption. This enables the creation of detailed charging demand models and node-level load prediction profiles.

1.1.1.5 Charging Station

A charging station is a road-side power supply point where electric vehicles (EVs) recharge their batteries. In simulation-based studies, charging stations are modeled as infrastructure elements . Those are connected to specific network nodes or road segments. Each station has certain characteristics, such as : charging capacity, availability, deployment strategy and charging session duration.

1.1.1.6 Simulation of Urban Mobility (SUMO)

SUMO (Simulation of Urban Mobility) [10] is an open-source, microscopic traffic simulation platform. It is used for modeling and analyzing transportation networks. SUMO supports user-defined road networks, custom vehicle flows, routing algorithms, and mobility patterns. Through its EV extension, Sumo can simulate battery discharge, energy consumption, regenerative braking and real-time charging events. Sumo provides a controlled environment for research on the EV ecosystem in Bangladesh. Here in SUMO urban traffic behavior can be simulated and the resulting charging demand can be measured without the need for large-scale real-world experiments.

1.1.1.7 Traffic Control Interface (TraCI)

The Traffic Control Interface (TraCI) [11] is SUMO's external programming interface. It allows real-time interaction between the simulation and external applications. TraCI enables dynamic control of vehicle characteristics, simulation variable extraction, traffic signal management and continuous logging of EV parameters such as SOC, speed, energy consumption and charging events. In EV load prediction research, TraCI plays an important role in automated data collection. It also integrates SUMO outputs with Python-based analytical models for prediction and visualization.

1.1.1.8 Power Consumption Analysis

Power consumption analysis refers to the study of how electric vehicles (EVs) use electricity across different driving conditions, speeds, accelerations and network paths. This includes quantifying energy consumption per route, per node and per unit distance [12]. Which helps researchers to understand which locations or routes are putting the most pressure on the grid. In urban environments like Khulna and Jessore, this type of analysis is really significant for identifying charging hotspots, peak demand periods and areas that need infrastructure reinforcement.

1.1.1.9 Load Forecasting Model

A load forecasting model is a computational method—often statistical or machine learning-based. It is used to predict future EV charging loads using past charging shortages, route flows , traffic congestion and charging station usage patterns. These models help predict upcoming power demand, optimize the placement of charging stations [7]. They also guide power suppliers in planning feeder capacity expansion. With the rapidly expanding use of electric vehicles in Bangladesh, such forecasts are essential to prevent transformer overloading and grid instability.

1.1.1.10 Urban EV Mobility Pattern

Urban electric vehicle mobility patterns refer to the characteristic movement behavior of electric vehicles within a city. It includes typical routes, peak congestion times, center-level activity , trip length and charging intervals. By understanding mobility patterns, researchers can link electric load distribution to traffic dynamics . In Bangladeshi cities —where electric vehicles like Easy Bikes, E-Rickshaws follow regular routes between markets, hospitals and government institutions. This analyzing mobility patterns becomes essential for accurate simulation modeling and load forecasting.

1.1.1.11 Queue Management and Promotion Logic

Queue management is an infrastructure-level process -which coordinates the flow of vehicles at a charging station. It distinguishes between active charging slots (where energy is transferred) and waiting slots (where vehicles are queuing for service). Promotion logic refers to the systematic process of moving a vehicle from a waiting state to a charging state as soon as a slot becomes available. This typically follows a first-in, first-out (FIFO) protocol to ensure fairness and simulation accuracy [13].

1.1.1.12 Rule-Based Recommendation System

A rule-based recommendation system is a decision-making algorithm -which uses some predefined "if-then" conditional statements to suggest the best optimal actions. In the case of EV charging, this system evaluates real-time telemetry such as a vehicle's SOC, distance to the nearest station and current queue length to recommend the most suitable charging locations [14]. It also balances the overall load of the network with the needs of each vehicle.

1.1.1.13 Heterogeneous EV Fleet

A heterogeneous EV fleet refers to a collection of different types of electric vehicles within a single simulation or study. It is characterized by different physical sizes, battery capacities and power requirements. In urban modeling in Bangladesh, this primarily includes a mix of easy bikes, electric rickshaws and electric vans . It is essential to take this diversity into account to capture the complex and uneven energy usage patterns. This is typical of real-world traffic.

1.1.1.14 Spatiotemporal Demand Pattern

Spatiotemporal demand pattern refers to the distribution of electrical load, across both space (geographical location of charging nodes) and time (simulation period or daily interval) . By analyzing these patterns, researchers and service providers can identify charging 'hotspots' and peak demand times. Wchich, enabling more precise grid reinforcement and infrastructure planning.

1.2 Problem Statement

The fast increasing of electric vehicles (EVs), specially in urban transportation systems, has created significant challenges in accurately prediction the charging load demand on the electricity grid. In Bangladesh, where electric three-wheelers such as e-bikes and electric rickshaws are widely used [5], the electricity distribution infrastructure is often already under stress during peak hours. The addition of EV charging intensifies this challenge due to the highly dynamic and nonlinear charging behavior . Which is influenced by factors such as -vehicle movement, battery state of charge , charging station availability and vehicle distribution at the system-level. A major limitation of existing research is the lackings of comprehensive and large scale real world datasets . That capture both vehicle and station level features, especially in developing areas. Besides, many conventional and statistical methods fail to effectively model the complex relationships between these variables. Where some advanced models can be computationally expensive and less intuitive for practical application in resource-limited environments . So, there is a critical need for a data-driven, effective and understandable approach to system level EV charging load demand prediction. This study addresses those challenges by using a simulation-based dataset that includes detailed system and vehicle characteristics . It implementes and evaluates multiple machine learning models to accurately predict the overall load demand (*total system load kW*) under changing system conditions.

1.3 Objectives

The primary objectives of this research are follows :

1. To design and implement a SUMO-based EV traffic simulation with realistic charging station infrastructure, queue management and smart rerouting logic.
2. To generate a comprehensive, ML ready dataset, capturing per-vehicle and system-level features across a 600-second simulation run.
3. To train and evaluate seven supervised machine learning and deep learning models —Linear Regression (LR), Decision Tree (DT), Random Forest (RF), Extreme Gradient Boosting (XGB), Artificial Neural Network (ANN), Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU) —for predicting the system-level EV charging load demand.
4. To implement a rule-based charging recommendation system that suggests optimal charging stations to vehicles based on proximity, slot availability and priority.
5. To analyze model performance using standard regression metrics (Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Coefficient of Determination (R^2)) and identify the most suitable approach for this domain.

1.4 Research Questions

This study seeks to answer the following research questions :

1. Can a SUMO-based microscopic simulation environment generate a comprehensive and realistic dataset for modeling EV charging load demand?

2. Which machine learning model provides the most accurate prediction of system-level EV charging load (`system_total_load_kW`)?

3. What are the most influential features affecting EV charging load demand in a multi-vehicle system?

4. How effectively can machine learning models capture the relationship between system-level and vehicle-level features for accurate load prediction?

5. How does the performance of linear models compare with tree-based models for EV load prediction?

6. How accurately can the trained models predict load demand across different system conditions?

1.5 Significance of the Study

The significance of this research lies in contributing to the growing challenge of managing electric vehicle (EV) charging demand through a data driven and scalable approach. With the rapid growth of electric vehicles, specially in developing regions like Bangladesh [5] —power distribution systems are increasingly facing unpredictable and highly variable load patterns due to unbalanced charging behavior. This study provides a realistic framework for accurately predicting system-level EV charging load (`system_total_load_kW`) using a comprehensive simulation-based dataset . It incorporates both vehicle-level dynamics and system-level operational characteristics. By applying and comparing multiple machine learning models, this study shows that advanced yet in computationally inexpensive techniques, particularly tree-based models- can effectively capture the complex nonlinear relationships inherent in the EV charging ecosystem. The study results highlight that load demand is largely driven by a few key system-level factors. Which is enabling stakeholders to better understand the underlying determinants of energy consumption. It has a direct impact on grid management, which includes improved demand prediction, optimal power distribution, improved planning of charging infrastructure and peak load stress mitigation on transformers and substations. Moreover, the use of understandable models and feature importance analysis ensures that the results are not only accurate but also interpretable. This is making them suitable for application in resource limited environments. Overall , this research bridges the gap between simulation-based data generation and real-world applicability. And this is providing a strong foundation for future research and practical applications in intelligent energy management systems for EV-integrated smart grids.

Chapter 2

Literature Review

Chapter 2: Literature Review

The research field related to EV charging load prediction and smart charging management encompasses multiple disciplines, including power systems engineering, transportation science and computer science . This chapter reviews the most relevant works in three thematic areas: (i) EV charging load modeling and pediction, (ii) simulation-based methods for EV data generation and (iii) machine learning methods for energy load forecasting.

2.1 EV Charging Load Modeling

Early approaches to EV charging load modeling dependent on probabilistic approaches, which used statistical distributions of arrival times, departure times and daily travel distances to generate artificial load profiles [15]. Sortomme et al. [6] showed that integrated charging strategies can significantly reduce distribution system losses compared to unregulated charging. He et al. [16] discussed the optimal placement of charging stations. Which showed that accessibility and driving range limitations play an important role in station usage patterns. More recent researches have adopted a data-driven approach. Chen et al. [12] integrated probabilistic load flow analysis with EV charging uncertainty to assess grid impact. This provides a basis for the feature engineering techniques. Which is also used in the present work. Ul-Haq et al. [15] modeled residential EV charging patterns and highlighted the importance of capturing spatial heterogeneity across charging locations.

2.2 Machine Learning for Load Forecasting

ML has become the dominant paradigm for short-term electricity load prediction. Zhang et al. [7] proposed a comprehensive deep learning framework for EV charging load forecasting. Which incorporated historical demand, time-of-day characteristics and weather data, achieving significant improvements over classical autoregressive models. Wang et al. [8] applied random forest (RF) regression in short term load forecasting at EV charging stations. That is demonstrating robustness against noisy sensor data and missing values. Gradient boosting methods have also achieved state-of-the-art results. Chen et al. [17] applied XGBoost to one-day EV load forecasting That is exploiting its regularization ability to prevent overfitting on small datasets. Tayeb and Hyndman [18] compared gradient boosting with conventional time series methods. And they found that tree-based methods consistently outperformed ARIMA models for nonlinear load patterns. Although Linear regression, simple remains a useful basis for load forecasting. Guo et al. [19] showed that linear models can capture the dominant trends in EV charging demand when the features are carefully designed, especially when polynomial and interaction terms are included. More recently, Siddiqui et al. [3] demonstrated that an integrated DeepBoost approach can further refine forecasting accuracy by combining deep feature extraction with boosting techniques. Similarly, Ran et al. [4] introduced a refined CNN-LSTM-AM model to specifically target multi-scale temporal dependencies in charging data, setting new benchmarks for sequential load prediction.

2.3 SUMO-Based EV Simulation

SUMO has been widely adopted for EV research because of its open-source nature, extensibility and support for battery models. Lopez et al. [10] describe the core capabilities of SUMO, including models of electric vehicle energy consumption, parking areas and a Traffic Control Interface (TraCI) API for external control. Wegener et al. [11] introduced the TraCI interface, that enables real time monitoring and manipulation of simulated vehicles. This makes it ideal for implementing adaptive charging management algorithms . Helbing and Triber [20] established the theoretical foundations for microscopic car-following models. That are the basis of SUMO's vehicle dynamics. The integration of these dynamics with electric powertrain models allows for accurate simulation of energy consumption patterns. And this directly affects the state of charge (SOC) trajectory and charging events.

2.4 Smart Charging and Recommendation Systems

Yilmaz and Crain [2] gave a comprehensive review of vehicle-to-grid technologies, identifying intelligent charging coordination as a key enabler of grid stability . Ding et al. [13] proposed a two-stage robust optimization for distributed energy management. That concept is related to the queue-aware routing logic implemented in this work. For EV charging coordination, Nguyen et al. [14] developed an adaptive dynamic programming method . Which showed that real-time SOC aware routing can significantly reduce the average waiting time. This motivates the rule based recommendation engine developed in the present study. Which selects stations based on SOC priority level and slot availability.

2.5 Research Gaps

Despite significant work in this area, several gaps remain. Most existing research relies on real-world datasets that are difficult to obtain, especially in the context of developing countries [5]. The creation of simulation based datasets using SUMO with integrated queue management and real-time recommendation systems has not been widely explored. Moreover, comparative benchmarking of multiple ML models on simulated EV load data is limited in the literature . This thesis addresses these shortcomings through an integrated simulation, data generation and ML prediction framework.

Chapter 3

Methodology

Chapter 3: Methodology

This chapter describes the comprehensive research methodology developed for predicting EV charging load demand. The approach integrates microscopic traffic simulation with advanced machine learning techniques to capture the complex, stochastically driven dynamics of urban energy consumption [10]. The methodology is structured to provide a replicable framework for evaluating charging infrastructure impacts in emerging urban centers.

3.1 Research Design

This research follows a quantitative, simulation-based experimental design centered on the "Monitor–Predict–Control" paradigm. Which is designed to bridge the gap between microscopic dynamics patterns and grid level electrical load - demand. As shown in Figure 3.1, the research design is performed through a systematic multi stage workflow. It begins with extensive **Data Collection** and **Environment Setup** on the simulation platform. Using this foundation, a **Custom Road Network Design** was performed to model the real characteristics of urban area of Bangladesh [5] (as for sample Khulna city's particular are is selected). Then **Traffic Light System Placement** and **Charging Station Placement** were strategically placed to simulate realistic infrastructural limitation and constraints. After the establishment of the physical environment , the approach shifts to modeling by determining **Vehicle Configuration in SUMO** [10] and **Vehicle Fleet Characterization**, representing the operational behavior of electric three-wheeled vehicles.

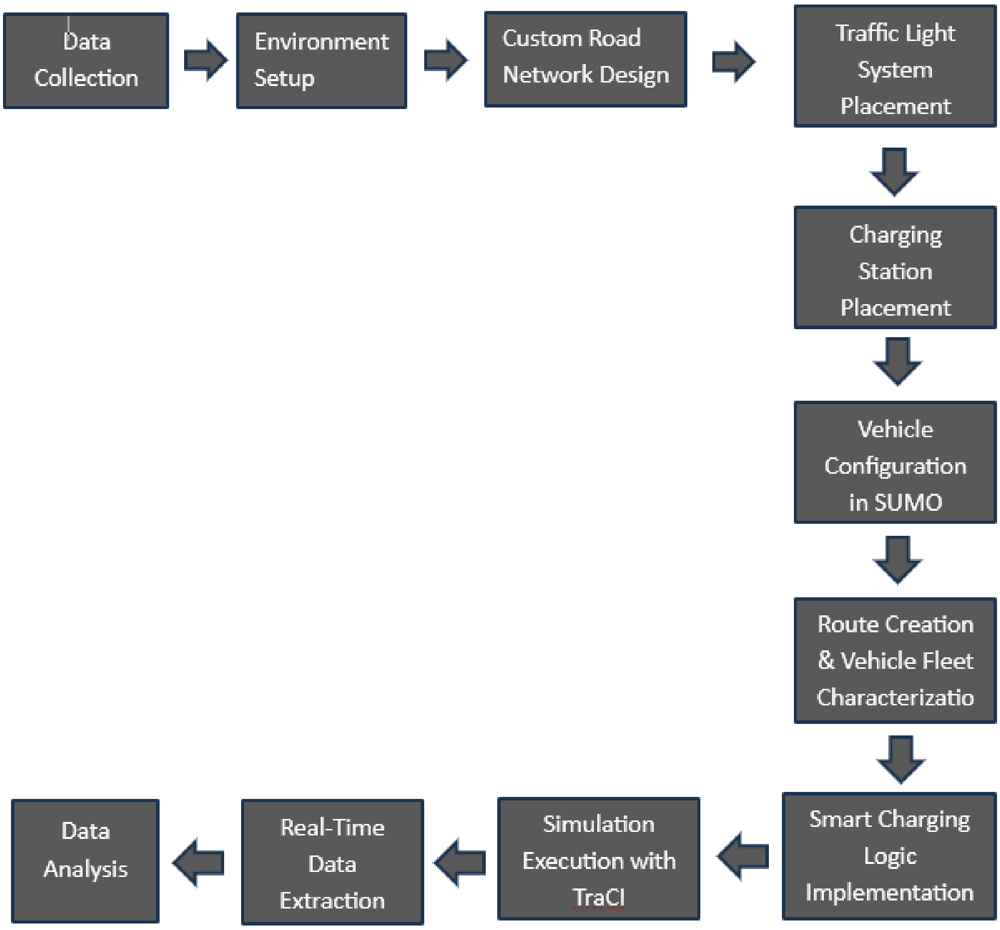

Figure 3.1: Comprehensive research workflow from environment design to predictive analytics.

To increase the system's realism and decision making capabilities , a queue management logic [13] has been implemented to manage vehicle congestion at charging stations, ensuring orderly access based on changing conditions such as queue length and availability. Additionally, a **Smart Charging Logic Implementation** is integrated into the simulation to guide vehicles to appropriate charging stations based on system state-dependent variables [2]. This introduces a rule based recommendation system and improves charging distribution across the network. The **Simulation Execution with TraCI** [11] is then run, enabling real-time interaction and control of the simulation process. During execution, a monitoring layer performs **Real-Time Data Extraction** at each timestep, capturing a variety of system-level and vehicle-level characteristics, including charging behavior, station utilization and load dynamics. The extracted data is further expanded to include insights related to charging recommendations, which is not the primary goal in this research-work. As it provides additional analytical value, this study can be more extended for further deep research of charging recommendation. Finally the collected dataset undergoes systematic **Data Analysis** and machine learning-based modeling to predict system-level load demand and evaluate the model's performance.

3.2 Data Collection

The data collection phase of this research was designed so that the simulation environment and subsequent machine learning models were based on realistic operational characteristics of electric three-wheeled vehicles commonly used in urban areas of Bangladesh [5]. Initially, detailed data was collected on three main categories of battery-powered vehicles : Easy Bikes, Electric Rickshaws and Electric Vans —which are known to be the main means of electric mobility in areas like Khulna.

The main objective of this stage was to capture both the physical and behavioral characteristics of these vehicles, so that the simulation could accurately reflect real world dynamics. We gathered vehicle specifications from a variety of sources—manufacturer information, publicly available technical documents and feedback from local drivers and operators. We thoroughly verified everything from these sources to ensure that the information was consistent and reliable.

The data included things like the size of the vehicles (length, width, height), their speed, how quickly they can accelerate or decelerate and all the key details of the power—such as battery size, charging capacity, energy consumption and mobility . We also looked at how these vehicles actually are operated in the real world : how drivers typically behave, their typical commute patterns and when and how they charge . As a result, we got a clear idea of exactly how these vehicles connect to charging stations in everyday life.

The data collection was not just focused on specific vehicle details. It also looked at broader contextual factors— such as how people use charging stations, how long charging-time typically are and how charging demand varies throughout the day. Those details were crucial to creating realistic simulation inputs and assembling a dataset that reflected both the details (such as specific vehicles) and the larger system as a whole.

After collecting and validating the data, we use it to determine vehicle types and create behavior rules in simulations, which lay the foundation for accurately generating and predicting load demand. Finally, this thorough data collection ensures that the research is not just on paper, it is consistent with what actually happens in electric mobility. As a result, the research survives in the real world and people can use its results.

3.3 Implementation

This section represents the detailed technical application of proposed research framework. Which includes the configuration of the simulation environment. The development of custom logical modules and the integration of data extraction and machine learning components. Main target of this implementation is to transform the conceptual "Monitor–Predict–Control" framework into a functional system capable of simulating realistic electric vehicles(EV) behavior. And this makes possible to generate high-quality data for load prediction.

3.3.1 Environment Setup

For this research-work, the simulation environment was carefully configured to support high reliably modeling of electric vehicle (EV) dynamics and charging behavior under controlled yet realistic conditions. All the experiments and tests were conducted on a 64-bit Windows 10-based workstation to ensure consistency, stability and efficient execution of both simulation and data processing tasks.

The main simulation platform used in this study is Simulation of Urban Mobility (SUMO) (Simulation of Urban Mobility, version 1.25.0) [10]. It is an open-source microscopic traffic simulator, globally familiar for its ability to model the dynamics of individual vehicles and its built-in support for electric mobility features, including battery management, energy consumption and charging station management [10].

We built the custom road network and simulation setup using SUMO's NetEdit, the graphical net-editor. It lets users design and tweak everything—roads, intersections, traffic signals, even where the charging stations are placed. With this tool, we were able to shape a real traffic scenario that reflects how things are laid out in urban areas like Khulna and Jashore city. For real-time interaction, we used the Traffic Control Interface (TraCI) (*Traffic Control Interface*) [11]. TraCI makes it possible for external programs to directly control the simulation, check system states and even modify vehicle behavior on the road.

We used Python 3.10.12 as the primary scripting language to control the simulation, collect data and integrate all analysis. With TraCI, we wrote Python scripts to handle each simulation step, set up queue management, enable smart charging features and collect real-time data while the simulation was running.

This data contains information about both individual vehicles and the entire system, such as speed, State of Charge (SOC), the number of charging vehicles, station occupancy and the total system load. We processed the resulting datasets using the `pandas` library and exported the refined data to structured formats like CSV or Excel for deeper analysis and machine learning modeling.

Overall, this integrated ecosystem ensures seamless integration between simulation execution and real-time monitoring while providing data-driven analytics. It provides a robust and flexible framework for EV charging ecosystem modeling, enabling the generation of high-quality datasets required for accurate load demand prediction and system-level performance evaluation.

3.3.2 Custom Road Network Design

In this research the created simulation environment represents a high-reliably, digitally constructed road network inspired by the structure of Khulna's arterial roads. The main purpose of this design is to realistically capture urban traffic patterns, vehicle interactions and their indirect impact on electric vehicles (EV) charging demand . The network is modeled as a connected graph, consisting of multiple nodes and edges, with a total estimated length of 8,654 meters. The network architecture was specifically designed to evaluate how vehicle density and road geometry influence energy consumption.

Each edge in the graph represents a real road segment, with specific characteristics such as the number of lanes, permitted speeds, and priority rules. By carefully adjusting these parameters, the simulation can simulate the different traffic conditions that exist in different parts of the city.

Table 3.1 provides a detailed description of these configurations. A detailed information about the connections between nodes, specific critical locations and operational constraints imposed on each part.

Table 3.1: Microscopic Road Network Configuration and Edge Parameters

Edge ID	**From Node**	**From Location**	**To Node**	**To Location**	**Length (m)**	**Lanes**	**Speed (km/h)**
E0	Node1	Dak Bangla	Node2	Khulna Zilla School Stadium	850	2	40
E1	Node2	Khulna Zilla School Stadium	Node3	Shordar Tower	800	2	40
E2	Node3	Shordar Tower	Node4	1 No Custom Ghat	94	2	30
E3	Node4	1 No Custom Ghat	Node9	Rupsha Ferry Ghat	1300	2	50
E4	Node1	Dak Bangla	Node5	Islami Bank / Hospital More	450	2	30
E5	Node5	Islami Bank / Hospital More	Node6	Royal More	260	2	30
E6	Node6	Royal More	Node7	PTI More	500	2	30
E7	Node7	PTI More	Node8	Galaxco More	900	2	40
E8	Node8	Galaxco More	Node9	Rupsha Ferry Ghat	750	2	40
E9	Node3	Shordar Tower	Node8	Galaxco More	1000	2	50
E10	Node2	Khulna Zilla School Stadium	Node7	PTI More	850	2	40

Note: All edges are bidirectional (2 lanes in each direction). Place names correspond to real-world Khulna City landmarks.

Important intersections and road segments are mapped to well-known urban landmarks such as Islami Bank Intersection, Royal Intersection, PTI Intersection and Rupsha Ferry Ghat. This ensures that the simulated environment closely reflects the geographical and infrastructural features of the real world. The entire network follows a left-hand traffic (LHS) system, which is standard in Bangladesh. The selection of Khulna City as the basis for this study is due to its rapidly developing urban infrastructure and the increasing popularity of electric three-wheelers. The landmarks mentioned are high-traffic hubs where charging demand is naturally expected to be concentrated. By anchoring the simulation to these real-world coordinates, we ensure that the results have practical relevance for local urban planning. Figure 3.2 highlights these arterial connections within the broader city map, illustrating the spatial relationship between the simulated edges and the actual urban grid.

Figure 3.2: OpenStreetMap (OSM) highlighted view of the Khulna City road network.

Figure 3.3 shows the custom microscopic layout built in SUMO using the `netedit` tool. To make traffic patterns feel real—especially the jams and car bunching—we've used static traffic lights at the busiest intersections. That's why the system puts lights at **Node 3 (Shordar Tower)** and **Node 8 (Galaxco More)**. Each light follows a fixed 90-second cycle with four phases: 42 seconds of green for the main direction, a quick 3-second yellow, then 42 seconds of green and another 3 seconds of yellow for the cross direction.

This cycle creates that classic stop-and-go traffic, which ends up making cars group together naturally. Clustering really matters here because it shapes how charging demand plays out. When a group of vehicles with low battery levels all hit the charging stations at once due to traffic delays, sudden spikes in load demand can occur. By adding controlled traffic interruptions like stoplights into the simulation, it's possible to capture those spikes more realistically. That's a key step to build and test models that predict load accurately.

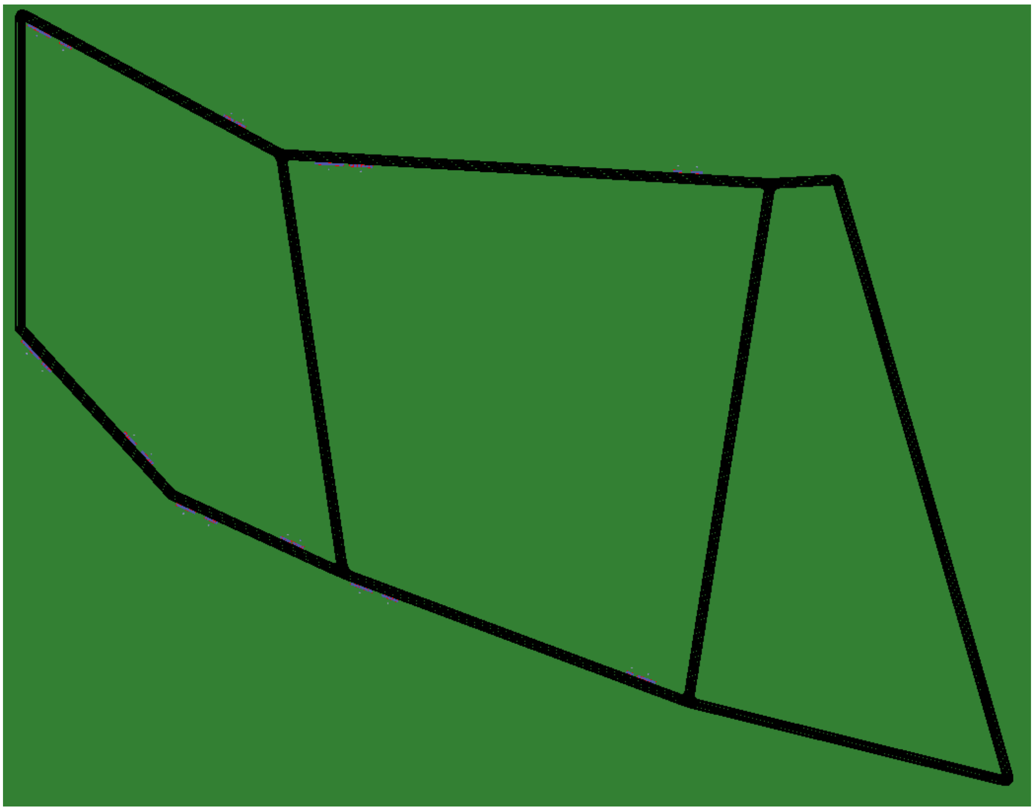

Figure 3.3: Custom microscopic road network in SUMO `netedit`.

3.3.3 Charging Station Placement

The strategic selection of charging locations is a critical factor for maintaining grid stability and ensuring vehicle accessibility across the urban network [16].

According to the actual data in the `Test1.chargingstations.add.xml` file, the charging stations are spread out on both the front and reverse sides of important road sections. Those road sections/edges are as: E0, E1, E5, E6, and E7. This setup covers the entire 8.6 km loop, so vehicles can always find a charging spot, no matter which way they're headed.

The stations don't all deliver the same power. Charging speeds go from 25 kW all the way up to 50 kW at the `pa_8_rev` station. This variation lets the simulation reflect a real-world mix of fast and standard chargers. Altogether, there are 62 charging slots and 49 spots for waiting vehicles, giving the simulation enough capacity to study sudden spikes in demand. For all the technical specifications of each station, check Table 3.2.

Table 3.2: Charging Station Technical Implementation Details

Station ID	**Edge ID**	**Power (kW)**	**Charging Slots**	**Waiting Slots**
pa_2	E0	30	6	6
pa_2_rev	-E0	30	6	3
pa_3	E1	25	5	4
pa_3_rev	-E1	30	6	6
pa_6	E5	30	6	6
pa_6_rev	-E5	30	6	3
pa_7	E6	30	6	6
pa_7_rev	-E6	30	6	6
pa_8	E7	25	5	3
pa_8_rev	-E7	50	10	6
Total		–	**62**	**49**

3.3.4 Vehicle Configuration in SUMO

To ensure the simulation reflects the real-world mobility landscape of urban Bangladesh, the vehicle fleet is meticulously configured with heterogeneous physical and electrical constants. We have modeled a total of 100 unique vehicle variants, distributed across three primary classes: 34 Easy Bikes, 33 E-Rickshaws, and 33 Electric Vans (E-Vans). This diversity is a critical design choice; by varying parameters like maximum battery capacity, vehicle mass, and speed factors, the simulation can accurately capture the stochastic nature of charging demand and the resulting load variations across a diverse urban fleet.

According to the configuration data established in the `Test1.rou.xml` file, each class is parameterized to reflect its unique operational characteristics. The **Easy Bike** variants are equipped with a substantial 12 kWh battery capacity and the largest frontal surface area (2.41 m^2), reflecting their role as the primary high-capacity mobility solution. In contrast, the **E-Rickshaws** are modeled with battery capacities ranging from 5.76 to 7.20 kWh and a taller profile of 1.8 m. Finally, the **E-Van** is implemented as a light-weight, low-profile vehicle with a 5.76 kWh battery and a height of only 0.8 m, significantly altering its aerodynamic drag and energy consumption profile.

All electric vehicles share a common emission class (`Energy/unknown`) and are configured with a moment of inertia of 0.010 $\mathrm{kg} \cdot \mathrm{m}^2$ and a stopping threshold of 0.1 m/s. To model realistic driver diversity, speed factors are distributed between 0.74 and 1.10, while maximum speeds are constrained between 40 and 50 km/h. These detailed specifications, summarized in Table 3.3, ensure that the simulated energy dynamics provide a high-fidelity dataset for load prediction.

Table 3.3: Cross-Class Vehicle Parameter Comparison (Averaged Values)

Parameter	**Easy Bike**	**E-Rickshaw**	**E-Van**
Body Dimensions (L×W×H)	2.5×1.5×1.4 m	2.3×1.0×1.8 m	2.4×1.3×0.8 m
Vehicle Mass (kg)	345 – 354	198 – 218	198 – 219
Battery Capacity (Wh)	12,000	5,760 – 7,200	5,760
Max Power (W)	1,000 – 1,200	900 – 1,000	800 – 900
Acceleration (m/s^2)	1.15 – 1.31	0.58 – 0.71	0.68 – 0.81
Deceleration (m/s^2)	2.45 – 2.58	2.48 – 2.67	2.58 – 2.77
Air Drag Coeff. (C_d)	0.727 – 0.760	0.804 – 0.910	0.754 – 0.860
Avg. Propulsion Eff.	0.890 – 0.896	0.830 – 0.834	0.820 – 0.826
Avg. Recup. Eff.	0.745 – 0.751	0.650 – 0.656	0.550 – 0.556
Const. Power Intake (W)	80 – 96	29 – 33	24 – 30
Front Surface Area (m^2)	2.41	1.51	1.01
Reaction Time (τ)	1.88 – 1.94 s	1.72 – 1.76 s	0.87 – 0.92 s
Visual Color (SUMO-GUI)	Magenta	Green	Blue

3.3.5 Route Creation & Vehicle Fleet Characterization

The construction of routes and the subsequent characterization of the vehicle fleet are essential for capturing the complex, multi-directional traffic dynamics of the study area. In this research, we have defined eight base routes—four forward and four reverse—which are meticulously mapped to the arterial edges of the Khulna city network to ensure bidirectional mobility across the 8.6 km loop. As summarized in Table 3.4, these routes ensure that the simulation captures bidirectional flow, allowing vehicles to traverse the corridor from multiple entry and exit points.

Table 3.4: Base Route Definitions and Edge Sequences in SUMO

Direction	**Route ID**	**Edge Sequence**
Forward	route1_forward	E0 → E1 → E2 → E3
Forward	route2_forward	E0 → E1 → E9 → E8
Forward	route3_forward	E4 → E5 → E6 → E7 → E8
Forward	route4_forward	E0 → E10 → E7 → E8
Reverse	route1_reverse	-E3 → -E2 → -E1 → -E0
Reverse	route2_reverse	-E8 → -E9 → -E1 → -E0
Reverse	route3_reverse	-E8 → -E7 → -E6 → -E5 → -E4
Reverse	route4_reverse	-E8 → -E7 → -E10 → -E0

Through assigning 100 distinct vehicle flows to these eight routes, fleet characterization further refines this system. Each flow is assigned a specific vehicle type (Easy-Bike, E-Rickshaw, or E-Van) and a measured number of departures ranging from 3 to 81 vehicles per hour. This custom diversity helps the simulation to mirror the real traffic in cities or urban areas across Bangladesh. In some areas, a lot of three-wheeled vehicles are seen on the roads, whereas other areas are not as busy. By mixing different departure times and entry points, the simulation mimics real-life, unpredictable traffic patterns. This setup helps test how well the load prediction models actually work. The summary of the fleet composition and color-coding for easy identification in SUMO-GUI in Table 3.5.

Table 3.5: EV Fleet Composition in SUMO Simulation

Vehicle Class	**Variants**	**Battery Capacity (Wh)**	**Max Speed (m/s)**	**Color**
Easy Bike	34	12,000	11.1 – 13.9	Magenta
E-Rickshaw	33	5,760 – 7,200	11.1 – 13.9	Green
E-Van	33	5,760	11.1 – 13.9	Blue
Total	**100**	–	–	–

3.3.6 Smart Charging Logic Implementation

The smart charging logic forms the operational core of the simulation framework. Which actually coordinates real-time decisions about when and where each electric vehicle will charge. The whole logic is encapsulated in the `SmartChargingStationMonitor` class. And this is implemented in the `load_demand_prediction.py` file.

This class communicates directly with the SUMO simulation engine via TraCI and simultaneously maintains a continuous understanding of the battery status of each vehicle and the number of vehicles at each station, every second, across a fleet of 100 vehicles. The monitor works with clear thresholds to manage when charging starts and stops. When the State of Charge (SOC) drops to 30%, charging kicks in. The goal is to reach a SOC of 70% by the end of each charging session. To avoid unrealistic simulation errors, the estimated charging duration is dynamically calculated. The calculation is based on energy shortages and station capacity. And the charging-duration is limited to between 30 and 300 seconds.

The monitor keeps tabs on all bidirectional charging stations and active EV agents using a few in-memory data structures. There's `station_charging_slots` and `station_waiting_queue`—which uses a `collections.deque`—for tracking station occupancy. On top of that, the `currently_charging` and `currently_waiting` dictionaries let it check real-time status for each EV. A comprehensive `vehicle_states` map stores per-vehicle metadata, such as charging start times, session SOC values, and routed station identities, ensuring continuous tracking and preventing duplicate rerouting commands across simulation timesteps.

When a vehicle's SOC falls below the 30% threshold, the system invokes a route-aware nearest-neighbor selection algorithm. Rather than relying on simple Euclidean distance, the monitor queries the engine via `traci.simulation.findRoute` for every candidate station. This allows the system to evaluate the actual network route length along the road graph, accounting for the unidirectional nature of lanes and city topology. Reachable stations are sorted by route length and evaluated according to a strict five-rule assignment hierarchy:

1. **Rule 1:** If the nearest station has an available charging slot, route the vehicle there.
2. **Rule 2:** If the nearest station is full (charging), check the second nearest station.
3. **Rule 3:** If the second nearest has a charging slot, route there.
4. **Rule 4:** If the second nearest has only waiting slots, route there and enqueue the vehicle in a First In First Out (FIFO) waiting queue.
5. **Rule 5:** If all candidate stations are full, do not reroute.

Once a target is determined, the vehicle is redirected using `traci.vehicle.setRoute` followed by a `traci.vehicle.setParkingAreaStop` command to register the stop in the simulation's schedule. For vehicles assigned to waiting slots, a **FIFO Queue Promotion Mechanism** is executed at every timestep. When a charging slot becomes available, the monitor retrieves the vehicle at the front of the station's deque using `popleft()`, resumes it via `traci.vehicle.resume()`, and reassigns it to the active charging area. This automated promotion ensures that waiting times are kept to a minimum and provides precise data on queuing delays. Upon session completion or vehicle departure from the network, the `handle_charging_complete` and `cleanup_departed_vehicle` methods record the final telemetry and clear the associated state maps to maintain simulation integrity.

3.3.7 Simulation Execution with TraCI

The simulation runs for 600 seconds, which gives plenty of time to see how vehicles interact from all directions and watch for any electrical load spikes in the area. The `load_demand_prediction.py` script handles everything—it kicks off the detailed environment in SUMO-GUI and sets up a fast connection with the TraCI API.

When the simulation runs, the loop moves forward one second at a time. Every time `traci.simulationStep()` kicks in, the `SmartChargingStationMonitor` jumps in to check up on all 100 cars. It doesn't just skim the surface—it tracks every little detail, like how the batteries are doing, where each vehicle has moved, and how fast they're going right at that moment.

When a vehicle's State of Charge (SOC) drops below 30%, the Python script jumps into action. It works out on the fly whether the vehicle needs rerouting or a charging stop, then sends the right commands right away. That quick response shifts the simulation from a fixed, predictable scene to something much more lively—one that can really capture all the messy, real-world charging patterns needed for accurate load forecasting. Once the simulation hits 600 seconds, the script wraps things up: it closes out the TraCI connection and gets busy turning all the collected vehicle data into the finished research dataset.

3.3.8 Real-Time Data Extraction

To create a solid dataset for the predictive models, we set up a real-time data extraction pipeline inside the simulation framework. This pipeline runs separately from the main mobility physics, which keeps the simulation stable and lets us grab high-resolution telemetry data. Every second, during each simulation step, the monitoring script checks all active electric vehicles through the TraCI interface and pulls their real-time performance metrics.

Capturing these power demand shifts at a high frequency—every second, actually—lets us spot quick, sharp changes that broader models just miss. Think of those sudden surges when a bunch of vehicles pile up at an intersection or when a new group arrives at a charging hub. Each record pulls together 25 features: some track what each car's up to (like speed, acceleration, state of charge), some focus on what's happening at the station itself (how many spots are full, how long the lines are), and others look at the bigger system.

Over a 600-second simulation, this approach collected 34,020 dynamic records—a pretty thorough set of data. At first, all this information gets stored in memory to avoid slowdowns from writing straight to disk. Once the simulation wraps up, everything is saved in a structured Comma-Separated Values (CSV) file. Table 3.6 breaks it all down, showing stats on things like vehicle statuses and power loads.

This dataset actually captures a wide range of real-world operating conditions. It covers everything from slow, low-traffic periods to those times when charging hubs are packed and congested. That mix gives the machine learning models really solid, realistic data to learn from.

Table 3.6: EV Charging Load Demand Dataset Summary

Property	**Value**
Total raw transactions	34,020
Number of features	25
Aggregated prediction samples	600
Unique vehicles	49

The dataset was then aggregated by time-step to calculate the total system load at every second, resulting in 600 high-fidelity samples for the prediction model development. This representative dataset ensures high-fidelity training for the subsequent machine learning models.

3.4 Machine Learning Application

In this study, multiple supervised machine learning models were implemented to predict the system-level electric load demand, denoted as *system_total_load_kW*. The problem is formulated as a regression task where the target variable is estimated based on a set of system-level and vehicle-level features [19].

3.4.1 Problem Formulation

The load prediction task can be mathematically expressed as:

$$y = f(X)$$

where:

- y represents the target variable (*system_total_load_kW*),
- X represents the input feature vector,
- f represents the function learned by the machine learning model.

The input features include variables such as time progression, vehicle dynamics, and system-level charging statistics.

3.4.2 Data Preprocessing

Before training the models, some preprocessing steps were implemented to ensure data quality and model efficiency:

- **Removal of Data Leakage:** The 'station_load_kW' feature was removed to prevent direct leakage into the target variable.
- **Removal of High-Cardinality Features:** Features such as:- vehicle_id, lane_id and current_station_id were excluded, as they were representing identifiers rather than generalizable patterns.
- **Feature-Target Separation:** The dataset was partitioned into input features-(X) and target variable-(y).

- **Validation Strategy (5-Fold Cross-Validation):** Instead of the usual random train test split, five fold cross validation method is applied to ensure strong performance evaluation. The dataset is divided into five equal subsets. In every iteration- four subsets are used for training and remaining subset for testing. The final metrics reported are the average values of five folds. Thus it minimizes bias from a single random split.

- **Temporal Aggregation and Sequencing:** To align vehicle level simulation logs with system level load requirements, two distinct data structures were utilized. For Classical ML and ANN models- the 34,020 raw transaction records were aggregated by timestep. This is resulting in 600 high fidelity samples representing the total system state per second. For Recurrent models (LSTM and GRU), the raw dataset was transformed into a sequential format using a sliding window of W = 5, allowing the architectures to capture temporal dependencies through five consecutive seconds of vehicle activity.

3.4.3 Deterministic Execution and Reproducibility

To ensure experimental accuracy and exactly reproducible results across different training cycles, a comprehensive deterministic seed block was implemented. This includes confining the global uncertainty of the environment by fixing seeds for the following: `PYTHONHASHSEED` (Environment level), `random` and `numpy` (Array and sequence level), `TensorFlow` and `TF_DETERMINISTIC_OPS` (Model and GPU hardware level). By reapplying these seeds within each iteration of the k-fold cross-validation and during the initial weight initialization of the neural networks- the variance from random weight assignment is eliminated. Which provides a stable benchmark for comparative analysis.

3.4.4 Feature Correlation Analysis

Feature correlation analysis helps to observe-which input features are most closely related to the target variable-*system_total_load_kW*. Here the Pearson correlation method measures how closely each feature value is consistent with the target.

The *system_charging_vehicles* has the strongest positive correlation with the target (about 0.996). In simple words, the main thing driving the system load is how many vehicles are charging at any moment. Time-based features like *timestep_sec* and *minute_of_hour* also show strong correlations, telling us that demand ramps up as time passes during a simulation.

Other broad system metrics—like *system_total_vehicles*, *system_moving_vehicles*, and *system_waiting_vehicles*—also tie closely to load demand. But it's wondering about things like *speed*, *acceleration*, or *status*, they barely move the needle. These features have weak or no correlation and don't really affect the total system load.

So, system-wide and time-related features play the biggest role in predicting load demand. Figure 3.4 presents the correlation matrix visualized as a heatmap.

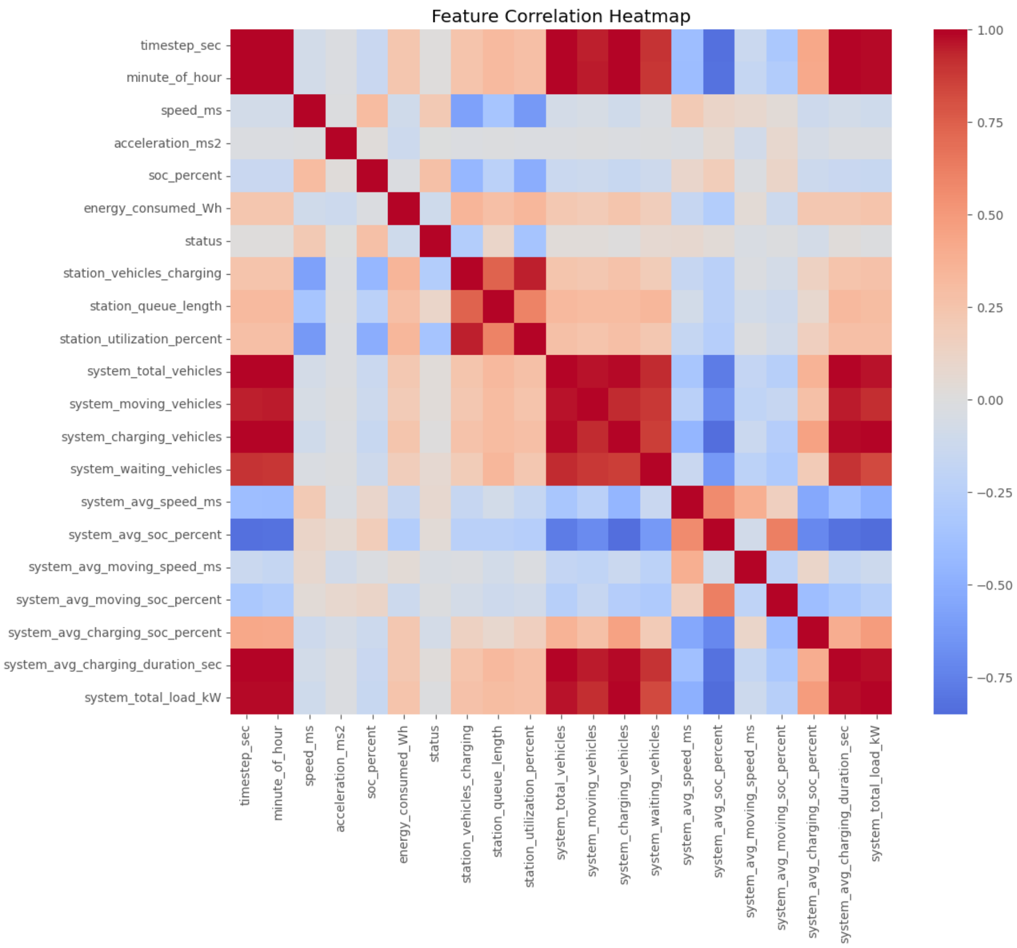

Figure 3.4: Pearson correlation matrix of the simulated EV charging dataset.

3.4.5 Feature Importance Analysis

We used the regressor to see which features really drive the prediction of *system_total_load_kW*. It turns out that *timestep_sec* is at the top, followed closely by *system_total_vehicles* and *system_charging_vehicles*. These three are the most important, as they reflect what is actually happening in the system—i.e., how active the systems are and how much charging is taking place.

Most of the other features are not that important. They don't add much, because those core features already explain almost everything the model needs to know. So this is clear that the model relies heavily on a handful of system-level features and that's enough to achieve high prediction accuracy. Figure 3.5 illustrates the importance distribution identified during the training phase.

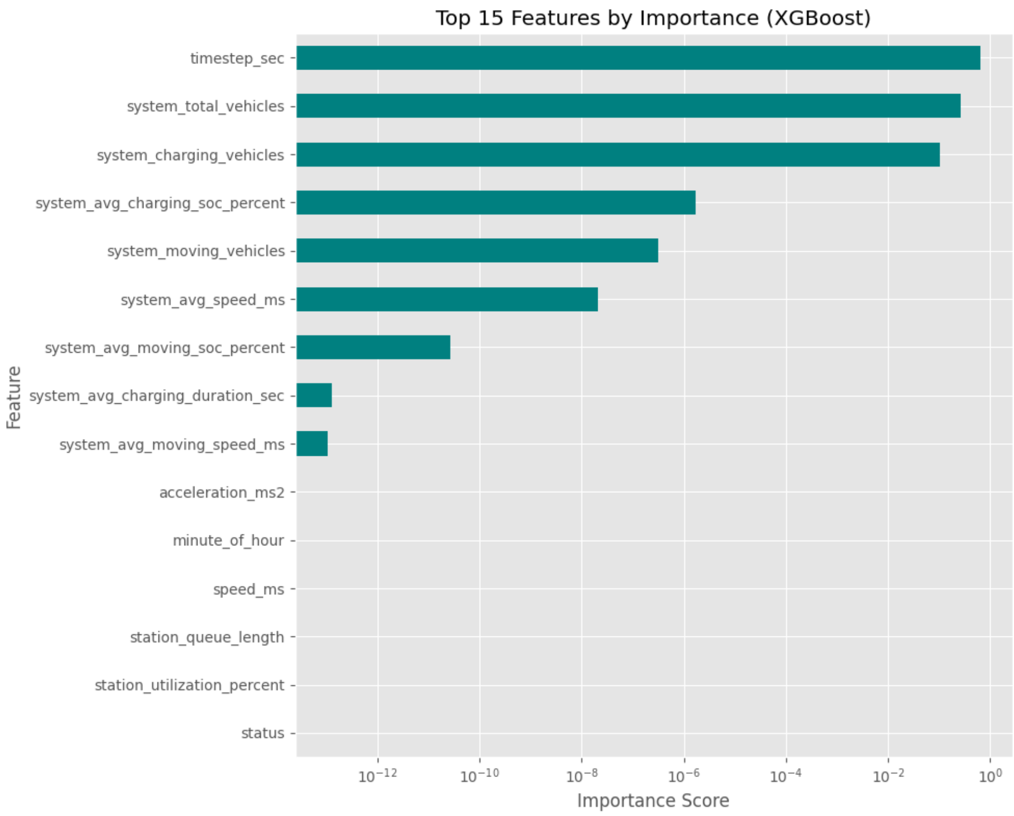

Figure 3.5: Feature importance ranking derived from the XGBoost model branches.

3.4.6 Implemented Machine Learning Models

After preprocessing complete- the dataset was split into training and test sets. 80% -data for training and remain 20% for testing. It ensures that the performance of model is evaluated on unknown data, testing its ability to generalize to new traffic scenarios. We implemented seven different regression-based models: Linear Regression; tree-based ensembles including Decision Tree, Random Forest, and ; and advanced deep learning architectures such as ANN, LSTM, and GRU. To maintain numerical stability during gradient descent, feature scaling was applied using the `MinMaxScaler` for Linear Regression and all neural networks, while tree-based models were trained on raw features as they are invariant to feature magnitude.

Once the training phase concluded, each model predicted the system load on the held-out test set. Evaluation was performed using Mean Absolute Error(MAE), Root Mean Squared Error ($RMSE$) and R^2 scores through five fold cross-validation to ensure robustness. Actual vs. Predicted scatter plots (Figures 3.6–3.12) shows the visualization of performance. Where closer proximity of data points to the diagonal line ($y = x$) signifies higher predictive precision.

(a) Linear Regression

Linear Regression shows the relationship between independent variables and the target as a simple weighted sum. In this research, we implemented **Ridge Regression** ($L2$ regularization). Where regularization strength of $\alpha = 100$. This choice was made because the system-level vehicle count has a near linear relationship with charging load. Unregularized regression risked achieving near-perfect training scores but failing to generalize. The regularization helps model to ignore noise in the aggregated simulation data. It also maintains stability across the five fold cross-validation. The prediction is calculated as:

$$y = w_1x_1 + w_2x_2 + \cdots + w_nx_n + b \tag{3.1}$$

where w represents the weights, x the features, and b the intercept.

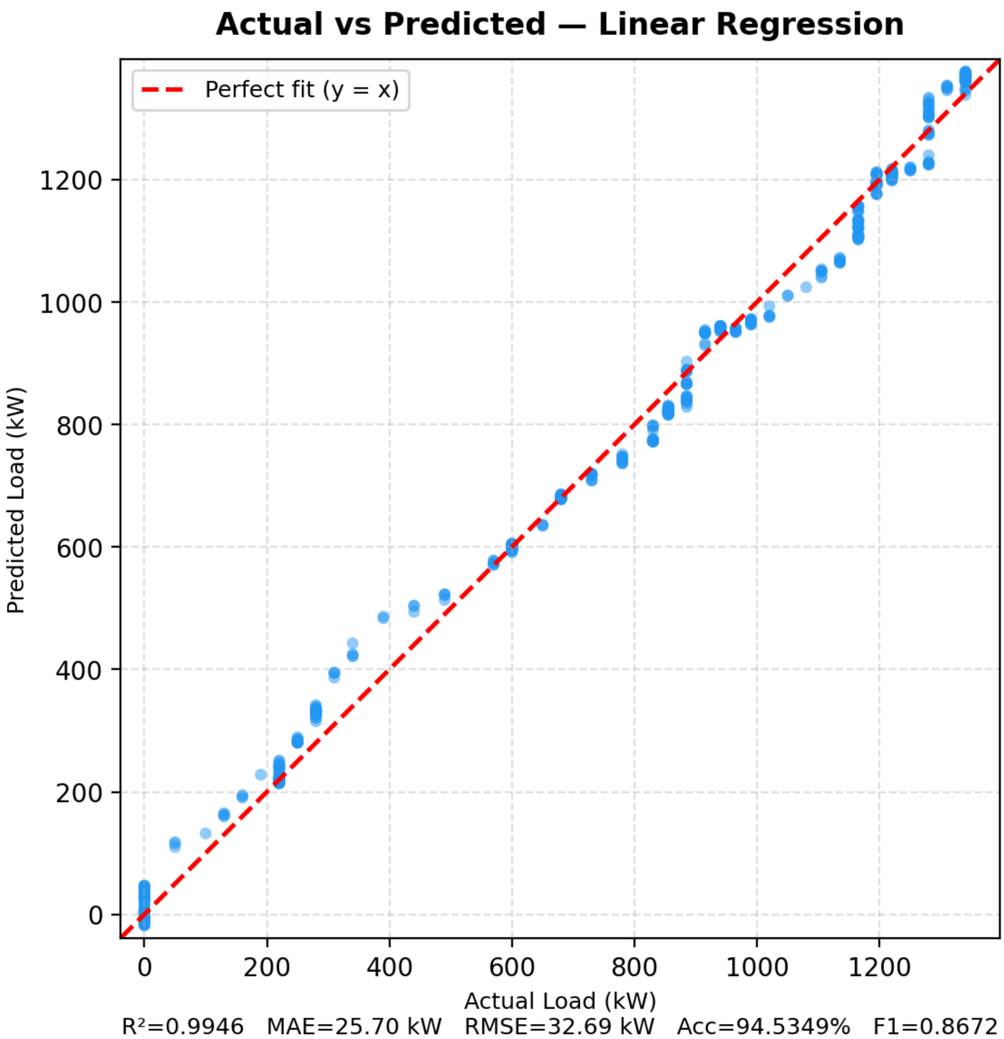

Figure 3.6: Actual vs. Predicted load for Linear Regression.

(b) Decision Tree

A Decision Tree splits the data into branches based on feature thresholds to minimize the prediction error(variance). To avoid individual data points-"memorizing" in our 600-sample dataset, we constrained the model to a maximum depth of *3* and required at least *30* samples per leaf node. This shallow architecture ensures that each terminal node captures at least 5% of the training data. That provides an honest and interpretable estimate of system load patterns.

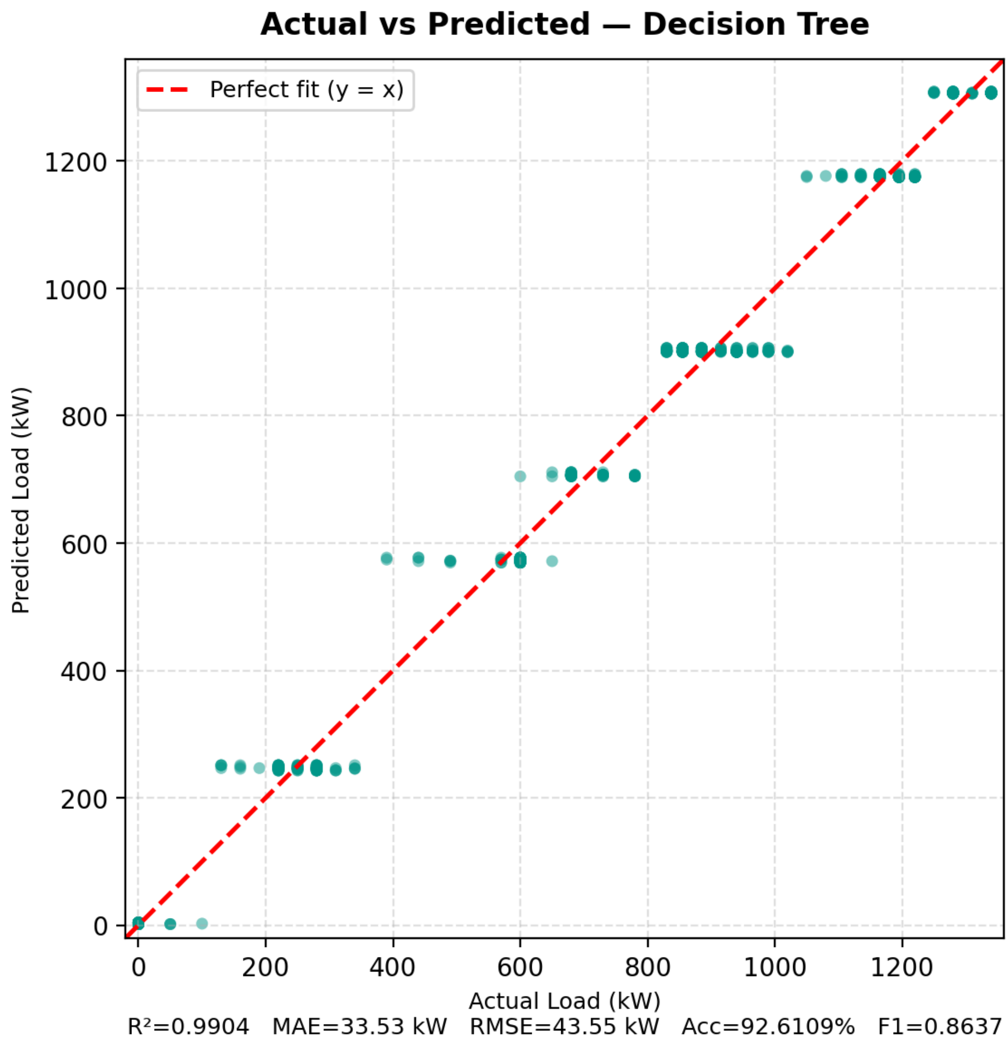

Figure 3.7: Actual vs. Predicted load for Decision Tree.

(c) Random Forest

Random Forest is an ensemble of multiple decision trees. Where each trained on a bootstrap sample of the data. To maintain robust generalization, we implemented a constrained ensemble of *50* estimators with a maximum depth of *3*. Besides, each split only considered *30%* of the available features (*max_features=0.3*). This heavy constraint prevents the forest from over fitting to simulation-specific artifacts. Through which it is ensured that the model identifies the fundamental drivers of load demand.

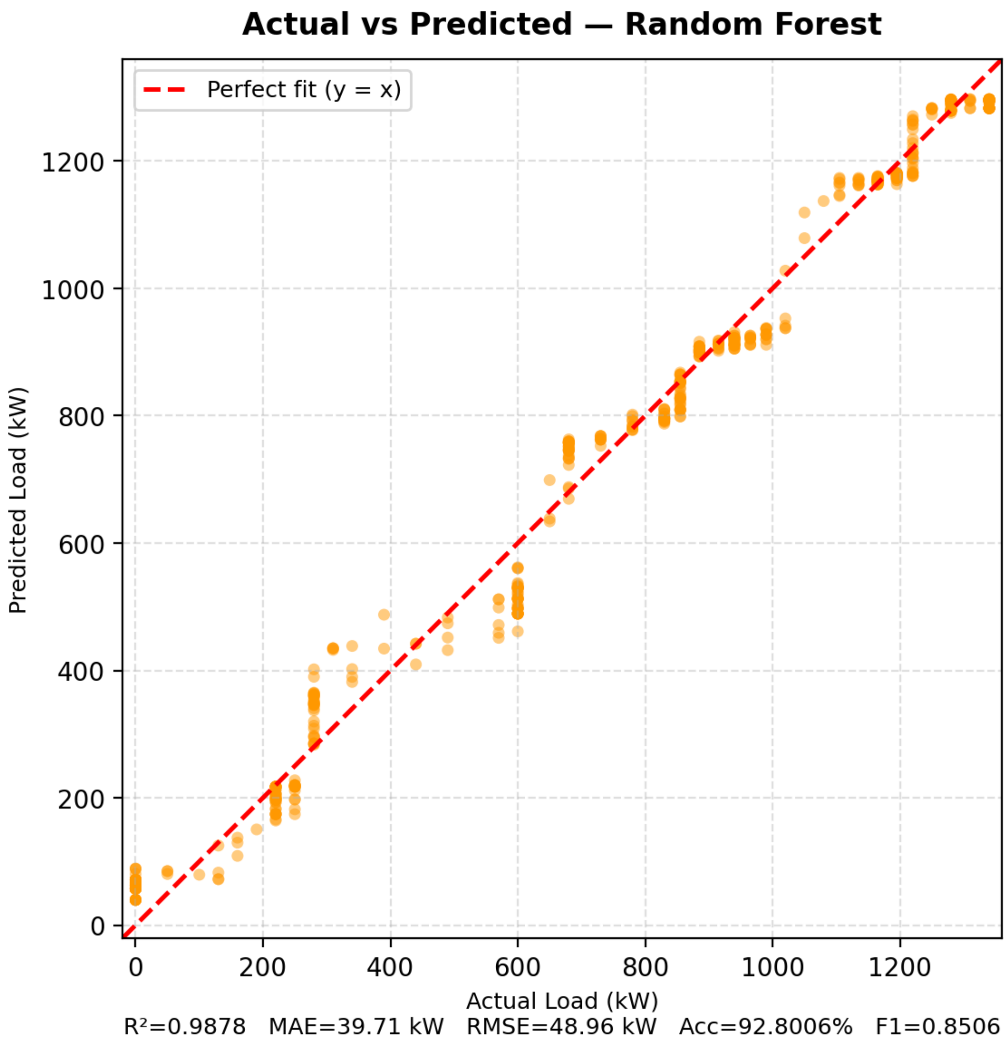

Figure 3.8: Actual vs. Predicted load for Random Forest.

(d) XGBoost

XGBoost is a powerful gradient boosting framework. It builds trees sequentially, with every new tree corrects the errors of the combined group. To apply this into our work, we used a low learning rate (*0.05*) and limited tree stumps (*max_depth=2*). We also implemented strong $L1$ and $L2$ regularization ($\alpha = 1, \lambda = 5$) and aggressive row subsampling (*0.55*). These parameters were optimized to prevent the boosting process from over converging on the small training set. Rather, it prioritizes long-term predictive stability over raw training accuracy.

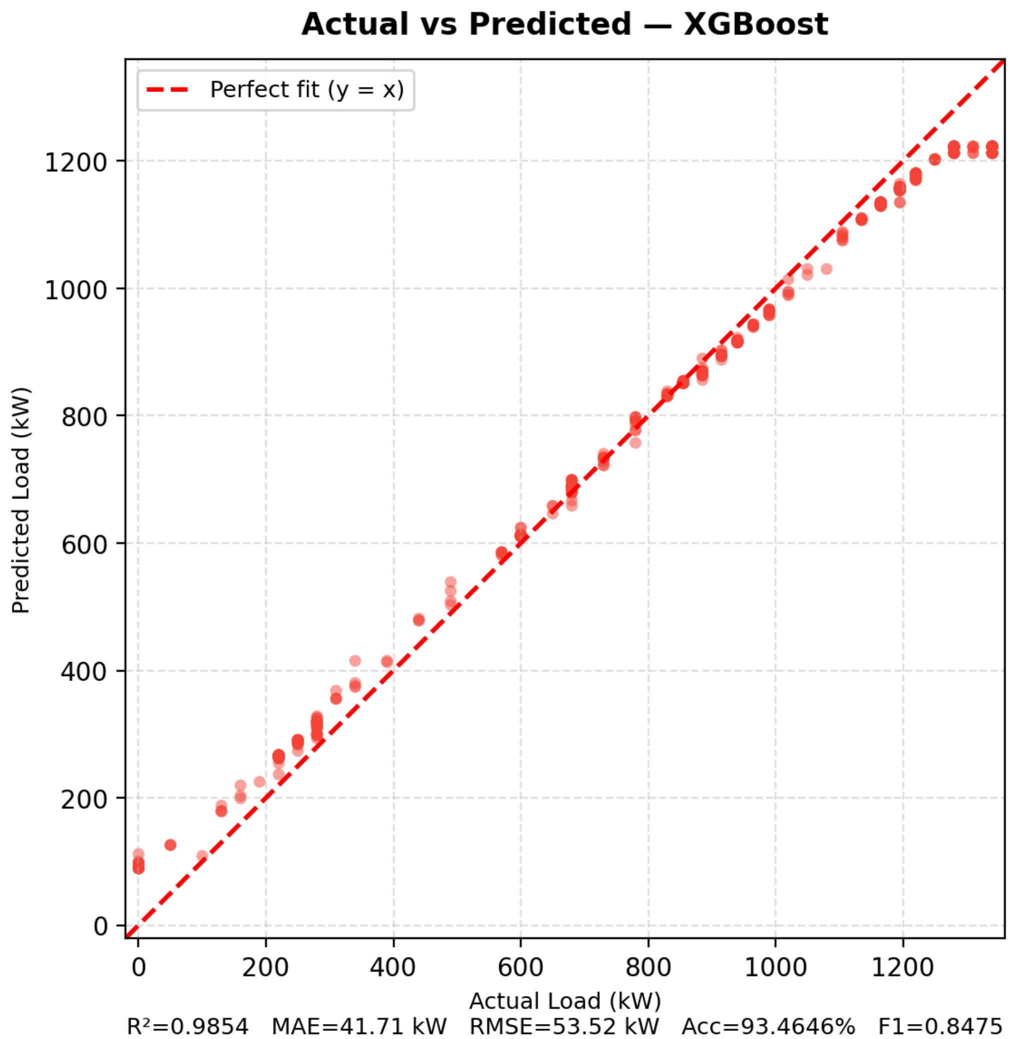

Figure 3.9: Actual vs. Predicted load for XGBoost.

(e) Artificial Neural Network (ANN)

In this research, the ANN was implemented as a Multi-Layer Perceptron(MLP). Which is consisting of 18 input features and followed by three hidden layers with 128, 64, and 32 neurons, respectively. Batch Normalization and Dropout (rate of 0.2) were integrated after the first hidden layer to enhance training stability and prevent overfitting. The model uses non-linear activation functions (ReLU), the Adam optimizer (learning rate of 0.001), and a batch size of 32 for efficient weight updates. As seen in Figure 3.10, the ANN proved to be the most powerful model in this study, achieving the lowest error rates and tracking the diagonal very precisely. Its ability to learn high-level representations from the feature-engineered dataset makes it exceptionally suitable for the dynamic environment of EV charging stations.

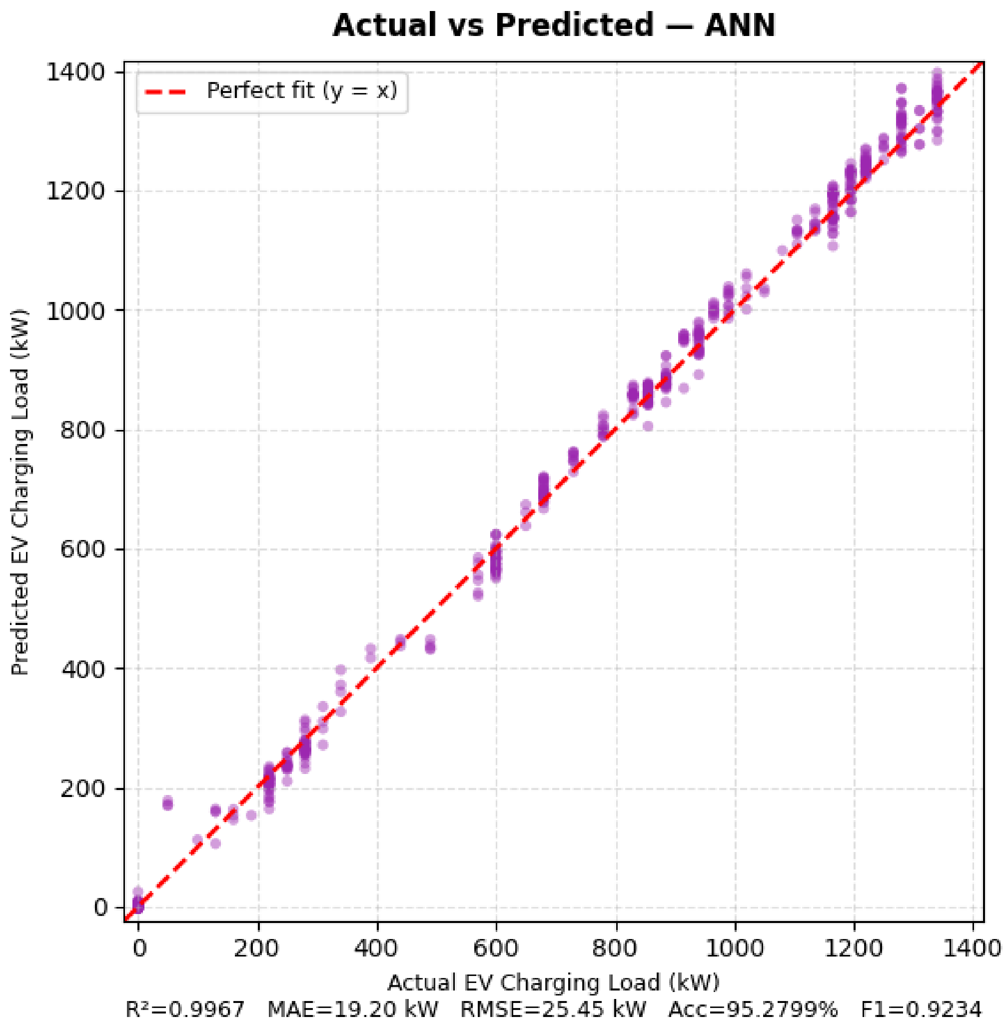

Figure 3.10: Actual vs. Predicted load for Artificial Neural Network.

(f) Long Short-Term Memory (LSTM)

In our setup, we used LSTM to see if the temporal sequence of the 600-second simulation could provide extra predictive power. The architecture consists of a 64-unit LSTM layer (with `return_sequences=True`), followed by a 0.2 Dropout layer, a 32-unit LSTM layer, another 0.2 Dropout, and a 16-neuron Dense layer with ReLU activation. A look back window of $W = 5$ was used. That means that the model considers the previous five seconds of system state to estimate the current load. Training was performed using the Adam optimizer and Mean Squared Error (*MSE*) loss, with an *EarlyStopping* callback (patience of 5) to restore the best weights and prevent overfitting. However, when the model performed well, it struggled slightly more than the ANN, which is reflected in the spread of points in Figure 3.11. This provides suggestion that- while time is important, the current system state (instantaneous features) might already carry enough information for dense networks to outperform purely sequential ones in this specific simulation scale.

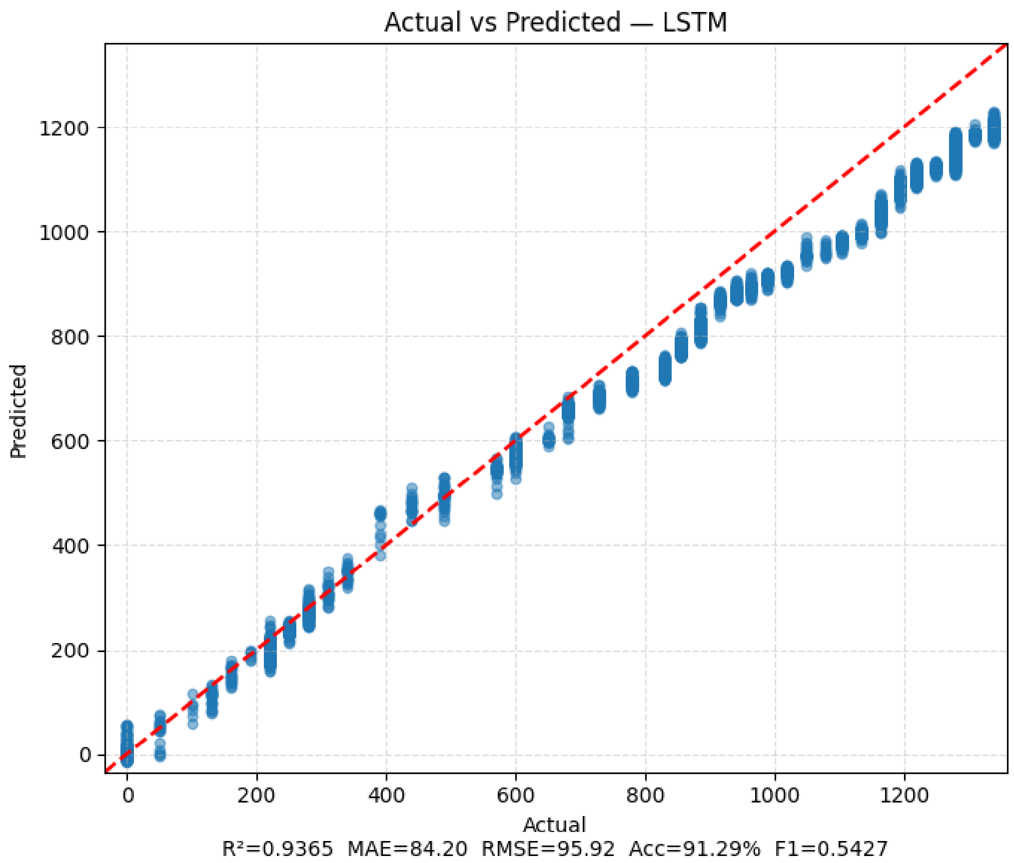

Figure 3.11: Actual vs. Predicted load for LSTM.

(g) Gated Recurrent Unit (GRU)

Gated Recurrent Unit (GRU) is a variation of the LSTM that simplifies the gating mechanism. It combines the forget and input gates into a single "update gate" and merges the cell state and hidden state. This makes GRUs computationally faster and often more efficient than LSTMs while still retaining the ability to process sequential information.

We implemented GRU as a faster alternative to LSTM to capture temporal dynamics, utilizing an identical structure consisting of stacked *64* and *32*-unit GRU layers with intervening *Dropout* (0.2). As with the LSTM architecture, a final *16*-neuron fully connected layer with ReLU activation is used to process the features before prediction. In our evaluations, its performance was similar to LSTM, proving that recurrent architectures can capture the general trend of the system load. However, as illustrated in the scatter plots, it fell behind the ANN and tree-based models in terms of raw precision on the test set. Nevertheless, it remains a strong candidate for real-time applications where processing speed is as critical as accuracy.

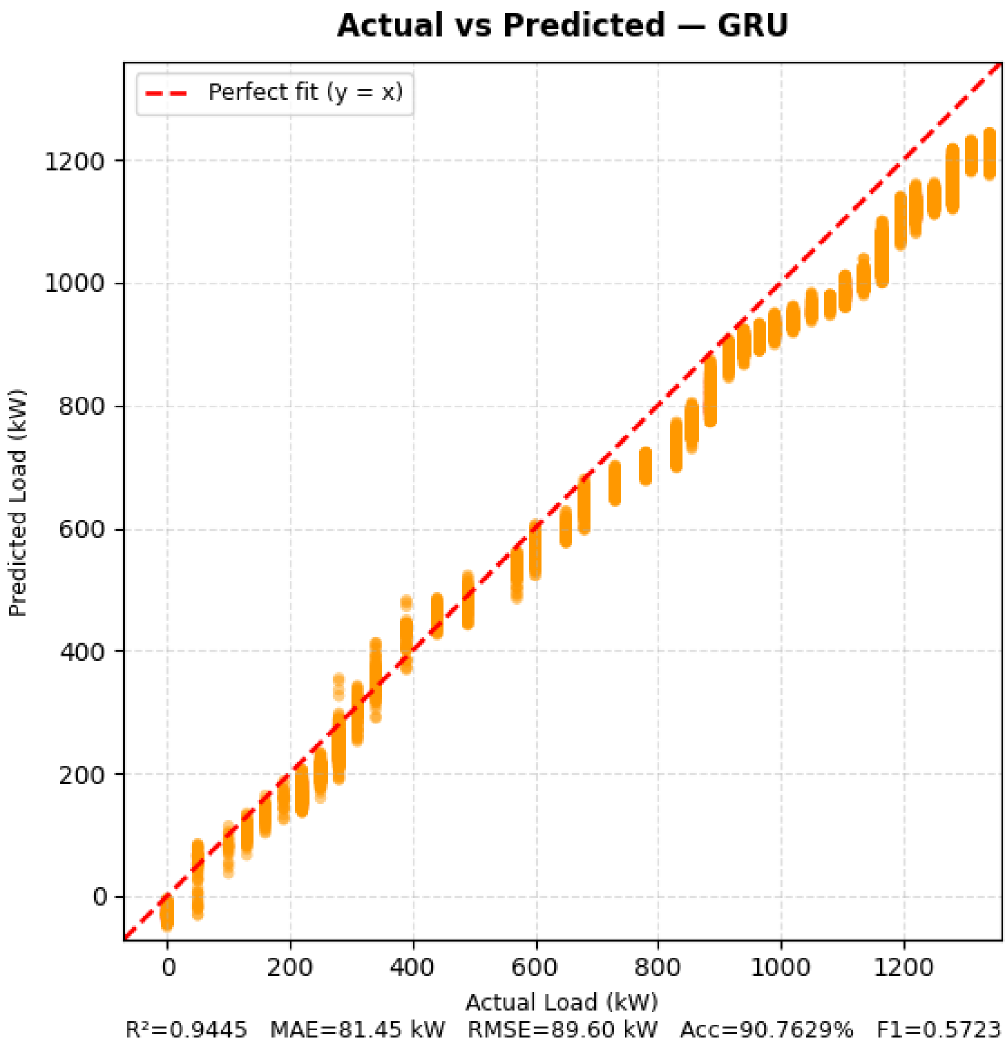

Figure 3.12: Actual vs. Predicted load for GRU.

Chapter 4

Results and Discussion

Chapter 4: Results and Discussion

This chapter presents the experimental results of the SUMO-based EV charging simulation and the Machine Learning (ML) load prediction models. Detailed discussion of the findings are as follows -

4.1 Results

The simulation really digs into how the electrical system and vehicle behaviors play out in this mixed EV network. Over 600 seconds, the model tracked everything at a fine level and collected a detailed dataset—34,020 records in all—catching every move and state change of the 49 vehicles in the fleet. The biggest takeaway is the **System Load Profile** shown in Figure 4.1. Right from the start, the load ramps up quickly as cars hit the road and begin their first round of charging.

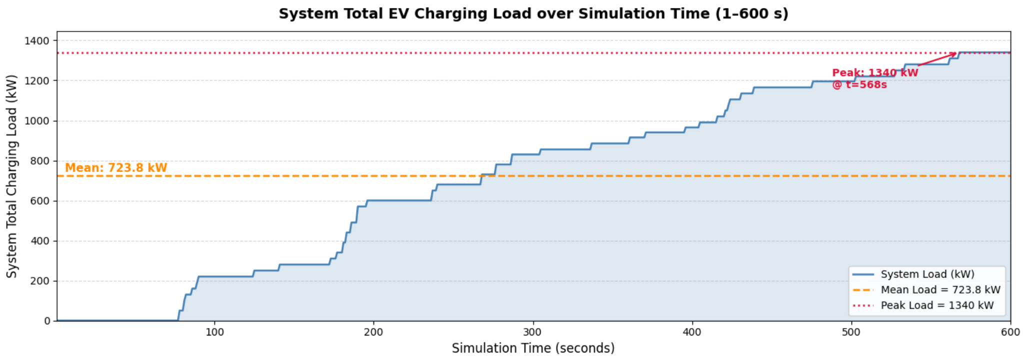

Figure 4.1: System total charging load over simulation time (1–600 seconds).

Power demand hit its highest point at 1,340 kW when the busiest charging stations—like `pa_8_rev`—were packed. On average, the system pulled about 723.82 kW, but the standard deviation was pretty high at 443.80 kW. That big swing shows how unpredictable things can get; sometimes a bunch of vehicles show up all at once at intersections or charging hubs and the load jumps, then drops right after as cars finish charging and leave.

Vehicle Status Distribution shows how vehicles spend their time: most of it goes to driving—about 58%. This means there's always a steady flow of cars that'll need charging soon. Around 35% of the total records show vehicles were actually charging, and that's the part putting real demand on the grid. Then there's the waiting status, which shows up in about 5% of the records. That's a sign that some charging stations are getting overcrowded, which then leads to delays and pushes the demand to other times.

Where all this happens isn't random, either. The layout of the roads really matters. Certain streets—especially those with heavy traffic, like pa_2 and pa_7—keep popping up as trouble spots where cars pile up and wait to charge. For charging demand, these high-traffic areas end up creating bottlenecks.

In addition to the core load demand forecasting framework, we added a charging recommendation system to make the simulation run more smoothly. This additional system provided 1,169 recommendations, directing vehicles away from crowded chargers and toward stations with sufficient space. Even though this "intelligent routing" wasn't the core of our predictive modeling, it made sure our dataset looked like it came from a smart, well-managed charging network—not just a jumble of random traffic. That way, when we trained our machine learning models, they picked up on patterns shaped by real system tweaks, making them stronger and more practical for real-world smart grid setups.

4.2 Discussion of Results

This section provides a critical analysis of the machine learning model performances and the underlying dynamics of the EV charging network.

4.2.1 Analysis and Interpretation

4.2.1.1 Evaluation Metrics

To evaluate the predictive performance and robustness of the machine learning models, seven distinct metrics were utilized. These provide a dual perspective on both predictive error and overall statistical fit. The metrics are mathematically defined as follows:

1. **Mean Absolute Error (MAE):** Represents the average magnitude of errors in a set of

predictions, without considering their direction.

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|$$

2. **Mean Squared Error (MSE):** Measures the average of the squares of the errors, giving higher weight to larger deviations.

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

3. **Root Mean Square Error (RMSE):** Represents the square root of the MSE, providing an error metric in the same units as the target variable.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

4. **Mean Absolute Percentage Error (MAPE):** Measures the accuracy as a percentage, representing the average absolute percent error for each time point.

$$\text{MAPE} = \frac{100\%}{n}\sum_{i=1}^{n} \left|\frac{y_i - \hat{y}_i}{y_i}\right|$$

5. **Accuracy (%):** In this regression context, accuracy is derived from the percentage error to represent how close the predictions are to the actual values.

$$\text{Accuracy} = 100 - \text{MAPE}$$

6. **Coefficient of Determination (R^2):** Indicates the proportion of the variance in the dependent variable that is predictable from the independent variables.

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}$$

7. **Weighted F1-Score:** To evaluate the model's ability to capture specific load levels (Low, Medium, High, Peak), the continuous predictions were categorized into bins, and the weighted F1-score was calculated to account for class imbalance.

$$\text{F1}_{\text{weighted}} = \sum_{j=1}^{L} \left(\frac{n_j}{n} \times \text{F1}_j \right)$$

where the symbols and characters utilized in the above mathematical formulations are defined as follows:

- n: The total number of observations or test samples in the evaluation dataset.
- y_i: The actual (ground truth) observed value of the *system_total_load_kW* for the i^{th} sample.
- $\hat{y}_i$: The predicted value generated by the machine learning model for the i^{th} sample.
- $\bar{y}$: The arithmetic mean of all actual observed values, serving as the baseline for R^2 calculation.
- L: The number of load categories (e.g., Low, Medium, High, Peak).
- n_j: The number of samples belonging to the j^{th} category.
- $\mathbf{F1}_j$: The F1-score calculated for the j^{th} category.
- $\sum$: The summation operator, representing the total sum across the specified range.
- $|\cdot|$: The absolute value operator, used to measure error magnitude regardless of sign.

4.2.1.2 Model Evaluation and Comparative Analysis

Performance results of the evaluated model's are given with details in Table 4.1. Which provides a quantitative benchmark through linear and ensemble-based architectures.

Table 4.1: ML Model Performance Comparison for System Load Prediction (5-Fold CV)

Model	**MAE**	**MSE**	**RMSE**	**MAPE (%)**	**R^2**	**Acc. (%)**	**F1-Wtd**
Linear Regression	25.702	1068.805	32.693	5.4650	0.9946	94.5349	0.8672
Decision Tree	33.530	1896.711	43.551	7.3890	0.9904	92.6109	0.8637
Random Forest	39.705	2397.463	48.964	7.1990	0.9878	92.8006	0.8506
XGBoost	41.712	2864.295	53.519	6.5350	0.9854	93.4646	0.8475
ANN (Proposed)	19.205	647.896	25.454	4.7200	0.9967	95.2799	0.9234
LSTM	84.202	9200.632	95.920	8.7093	0.9365	91.2907	0.5427
GRU	81.451	8028.416	89.495	9.2371	0.9444	90.7629	0.5723

The combined results presented in Table 4.1. It gives a comprehensive quantitative overview of the performance of each model architecture—linear, tree-based and recurrent—tested on the simulation dataset. By including seven distinct metrics, the table offers a dual perspective on both predictive error (MAE, MSE, RMSE, MAPE) and overall statistical fit (R^2, Accuracy, F1 Score). These numbers serve as the baseline for a more detailed comparative analysis of model stability and precision.

A detailed breakdown of the model error magnitudes is provided in Figure 4.2, which highlights the Mean Absolute Error (MAE) and Mean Squared Error (MSE) scores.

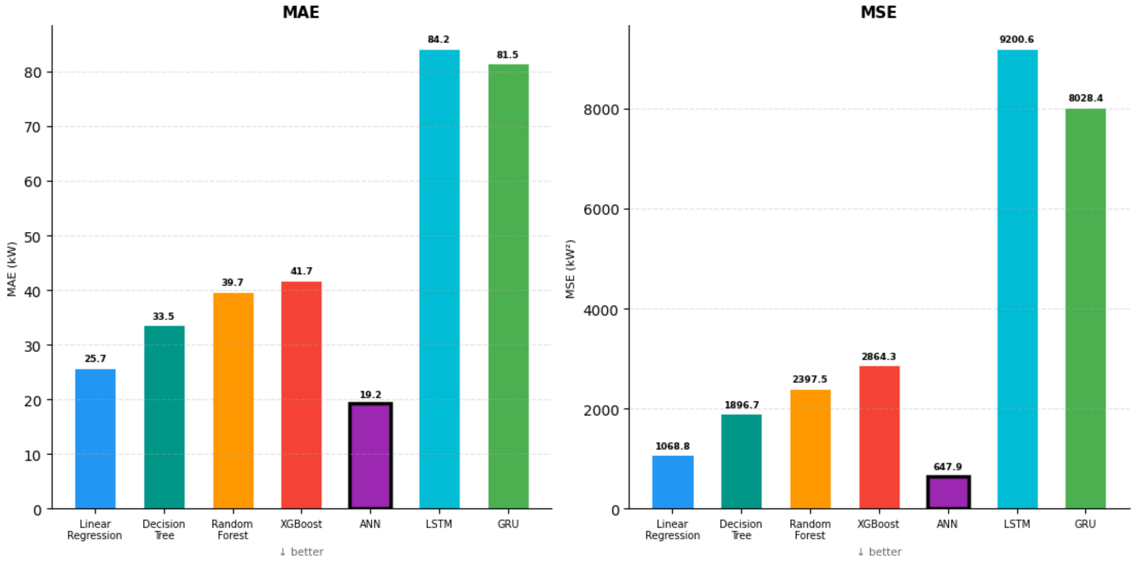

Figure 4.2: Mean Absolute Error (MAE) and Mean Squared Error (MSE) comparison across models.

By observing the error magnitudes, the **Artificial Neural Network (ANN)** stands out as the most precise model, racking up MAE values of only 19.21 kW and demonstrating the lowest variance in squared error (MSE). The **Linear Regression** baseline also performed exceptionally well ($MAE \approx 25.70$ kW), actually outperforming the tree-based ensemble models like Random Forest and XGBoost in terms of regression error on this specific dataset. This indicates that a significant portion of the EV charging load demand is driven by linear relationships, specifically the number of actively charging vehicles.

Further evaluation of model consistency and relative error is shown in Figure 4.3, which compares the Root Mean Square Error ($RMSE$) and Mean Absolute Percentage Error ($MAPE$).

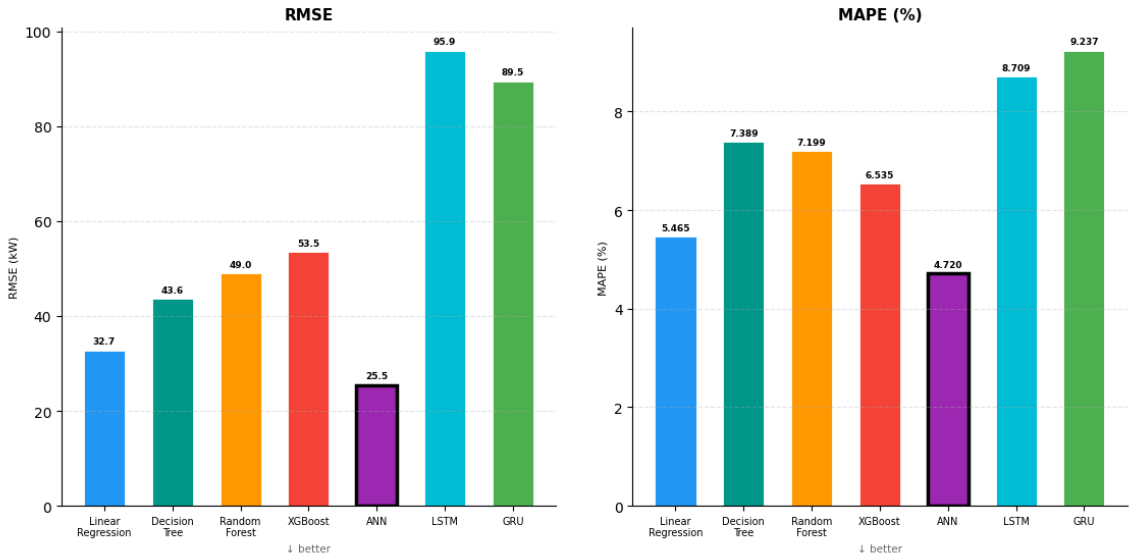

Figure 4.3: Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE) comparison.

The tree-based models like Decision Tree, Random Forest and XGBoost still held up strong with R^2 scores above 0.98. However it shows that with more rigorous cross-validation, we're getting a more realistic picture of how these models handle variance. On the other hand, the recurrent models (LSTM and GRU) struggled more. LSTM, in particular, had the highest error rates ($MAE \approx 84.20$ kW and $RMSE \approx 95.92$ kW). What this tells us is that while these models are great for patterns that stretch over time, the current "shuffled" state of the simulation data might be better suited for dense or linear architectures. Figure 4.4 shows the comparative distribution of R^2 scores of the models.

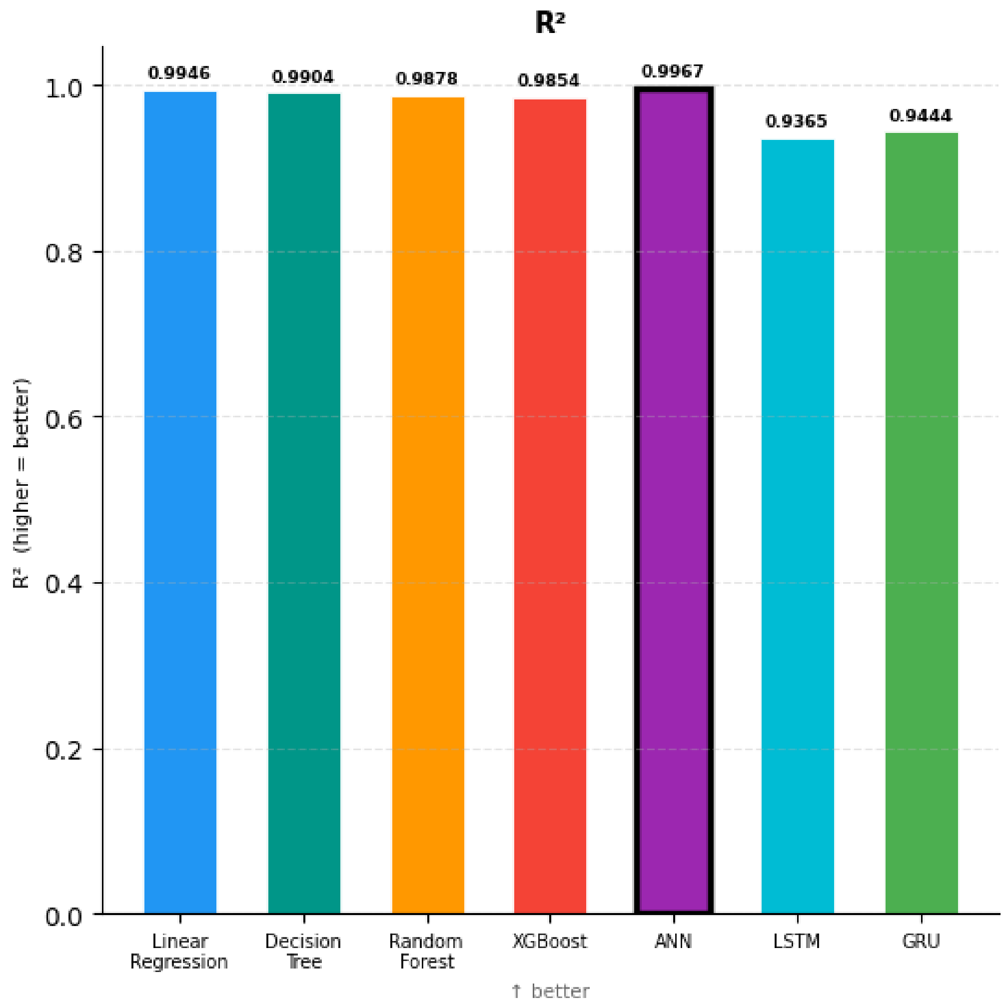

Figure 4.4: Coefficient of Determination (R^2) Score distribution for supervised machine learning and deep learning models.

Additionally, Figure 4.5 shows the classification-based performance scores. ANN is again at the top with an accuracy rate of 95.28% and a weighted F1-score of 0.9234. It handles the preciseness of the data better than anything else.

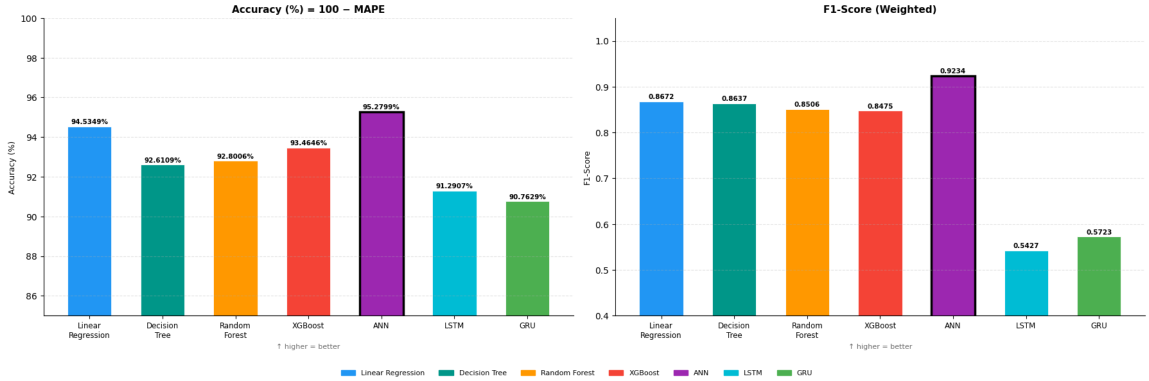

Figure 4.5: Comparative performance distribution for Accuracy and Weighted F1-Score.

4.3 Implications

The results of this study provide important insights into load forecasting, smart charging infrastructure and data driven urban energy management based on electric vehicles. The implications extend to grid management, urban planning and intelligent system design. Specially in the context of developing cities like Khulna.

4.3.1 Grid and Charging Network Management

The near-perfect prediction accuracy of machine learning (ML) models ($R^2 \approx 0.9967$) shows that it is possible to reliably predict system-level EV charging loads using data from simulations. This enables power grid operators to adopt a **predictive management approach**, which can anticipate demand before peak congestion occurs. This methodology is consistent with recent findings in integrated forecasting frameworks that emphasize the synergy between simulation and deep learning to stabilize urban grids [3]. Moreover, the simulation also shows that using smart charging logic with queue management spreads out charging demand across stations. This helps cut down on local overload. It's clear that blending real-time control strategies with predictive models really boosts grid stability and makes everything run smoother.

4.3.2 Data-Driven Feature Insights

By looking at feature importance and correlation analysis, it's clear that EV load demand mainly depends on just a few big-picture variables—especially charging activity and when those activities happen over time. It's not each car's tiny details, but the overall behavior of the system that is really driving the load. In practical terms, this means there is no need to track every little thing about every vehicle. By focusing on a handful of key features, it is possible to monitor EV load demand accurately in the real world. Details like state of charge, speed, or how the system moves around help set the scene, but they aren't pulling the biggest levers. This makes life much easier for anyone running real-world deployments; by zeroing in on just a few important metrics, it is still possible to get sharp, reliable predictions without having to collect a mountain of data.

4.3.3 Urban Infrastructure Planning

The simulation clearly shows how important it is to choose carefully where to place charging stations in a city. Busy intersections and popular spots draw the most vehicles—they basically suck up energy and can turn into serious bottlenecks. If too many cars pile up there, it doesn't just slow down traffic; it also puts extra strain on the power grid. With this framework, urban planners can run virtual stress tests on charging infrastructure before anything gets built. That takes a lot of the guesswork out of it, lowers investment risk, and helps them pick the best spots—especially in cities like Khulna.

4.3.4 Simulation as a Digital Twin Framework

The SUMO simulation environment works like a digital twin of city traffic and charging systems. The simulation can give a clear view of how vehicles move around and how they charge in real time. It also helps to build high-quality datasets for machine learning and to run all sorts of traffic and charging scenarios—with no risk and no actual road needed. That's a huge help in places where there is not much real-world EV data. Simulation closes the gap, making research and planning a lot more practical and scalable.

4.3.5 Machine Learning for Load Forecasting

When comparing the machine learning models, it's clear that tree-based methods like Decision Trees, Random Forest, and especially leave linear models in the dust. stands out as it nails that sweet spot between accuracy and generalization. The accuracy numbers aren't just high—they're almost off the charts—which means the data has some strong, predictable patterns. All this just proves one thing: machine learning is a powerful tool for tackling structured EV load prediction. Broader research trends similarly suggest that while tree-based models excel in structured settings, hybrid architectures such as CNN-LSTM-AM are increasingly necessary to capture more complex, multi-scale temporal dependencies [4]. Consequently, these models can be directly integrated with smart grid systems to obtain reliable and real-time forecasts.

4.3.6 Roadmap for Smart and Sustainable Electrification

This research lays out a real, workable plan for bringing electric vehicles into growing cities. Simulation is used to create real-world data, building machine learning models to make predictions and designing smart charging systems with queue management. All of this comes together in a **Monitor–Predict–Control** framework, which can help cities actually manage EV networks day-to-day. With this setup, institutions and policymakers get the tools to build solid, data-driven policies for transportation and energy. Consequently, cities can welcome more EVs without putting the grid at risk.

Chapter 5

Conclusion

Chapter 5: Conclusion

This thesis introduces a complete framework for predicting EV charging demand. We combined a detailed traffic simulation with supervised machine learning to make it work. To set things up, we used SUMO and the TraCI Python interface—basically, they let us manage and simulate a virtual road network. In our model, there are 10 adjustable charging stations, several Electric Vehicles and a smart system that handles queues and offers real-time charging recommendations.

The simulation generated a dataset with 34,020 records and 25 features. Which is tracking both individual vehicles and the whole system over 600 seconds. We tested seven machine learning models, ranging from Linear Regression to advanced Artificial Neural Networks (ANN). The ANN-model achieved the highest precision with an R^2 of 0.9967. Interestingly, Linear Regression also scored a very high R^2 of 0.9946. Where tree-based and recurrent models (LSTM and GRU) also provided reliable but slightly less accurate load forecasts.

5.1 Concluding Remarks

Here's what stands out in this study:

1. We built a powerful SUMO-based microscopic simulation [10] just for Bangladesh's diverse traffic mix—think Easy Bikes, Electric Rickshaws, and Electric Vans, all accurately modeled.

2. There is a solid dataset for machine learning, which contains 34,020 records and 25 dynamic features. It's more than enough to really dig into energy usage and build accurate load predictions.

3. We put seven regression models head-to-head, including deep learning architectures, and the Artificial Neural Network (ANN) emerged as the most suitable model for short-term charging load forecasting in this simulated environment.

4. We created a rule-based smart charging recommendation system, showing it's possible to reroute vehicles in real time based on SOC to avoid grid congestion in certain areas.

Our results show simulation-based data works really well when real-world EV charging datasets are hard to find—especially in fast-changing places like Bangladesh. This work gives city planners and power companies a data-driven way forward as transport electrifies, helping them keep the grid steady with strong, accurate forecasting tools.

5.2 Limitations

We got solid results predicting EV charging loads and running simulations, but there are still some real limitations here. It's important to keep these in mind when looking at what we found and thinking about how we can do better next time.

Dependence on Simulation Data: For this research-work, the dataset was built using a SUMO based simulation instead of real traffic data. We have made great efforts to make the simulation as close to reality as possible. Through which, we have tried to reflect real traffic patterns and the behavior of electric vehicles. Still, simulations can only go so far. There are always those unpredictable things— how people actually drive, sudden traffic jams or drivers charging their cars at odd times —that are tough to capture on a computer. Because of that, the model probably looks a bit better in testing than it would in the real world.

Limited Feature Diversity: The dataset (from the SUMO-simulation) covers a range of system-level and vehicle-level features. But it misses some important real-world details. Things like : weather—temperature or rain—aren't included. There is nothing about how people behave when charging like- their preferences or sense of urgency . Changes in electricity prices or tariffs aren't factored in and the model doesn't account for sudden infrastructure failures. Without these elements, the model doesn't tell the whole story.

Static Infrastructure Assumptions: The simulation sticks to one setup: charging stations never move, their capacity stays the same, and the network layout doesn't change. But that's not really how things play out in the real world. Charging infrastructure shifts—new stations pop up, old ones get shut down for maintenance, and everything from network design to capacity can shift around. These changes can throw off predictions and affect how the load gets spread out.

5.3 Recommendations for Further Research

For further extension of the scope and applicability of this research, several recommendations for future work are identified:

Integration of Real-World Data: Going forward, researchers need to lean on real-world EV charging and traffic data—to make their work hold up outside the lab. SUMO simulations are great for control and detail, but without much realistic data, they miss a lot. Real traffic is messy. People drive unpredictably, infrastructure isn't always up to speed, and demand can swing wildly in ways a simulator might gloss over. Mixing in data from actual charging stations, smart meters, or transport authorities will let future models learn from the real thing. That makes predictions tougher but also way more dependable and relevant. This shift will help these predictive models actually work when plugged into real-world smart grid systems, especially in places like Bangladesh, where data-driven city planning is just starting to catch on.

Advanced Deep Learning Architectures: Although this study integrated fundamental deep learning models like ANN, LSTM and GRU, there is still some of space to push things further. Next steps could look at even more refined hybrid architectures such as CNN-LSTM-AM [4]. These models are built to pick up on both spatial and multi-scale temporal patterns. Which actually matters a lot for EV charging since demand shifts throughout the day. By feeding these models with historical data and letting deep learning do the heavy lifting, future researchers can paint an even clearer picture of how load demand changes over time.

Intelligent Charging Optimization: In this study rule based smart charging system is implemented. This gives a basic method for distributing charging demand among multiple stations.

Anyway, it can be significantly enhanced by using intelligent optimization techniques. Future research could include- advanced methods like reinforcement learning, genetic algorithms or multi-objective optimization to enable adaptive and autonomous decision-making . These kind of systems can assign dynamically -vehicles to optimal charging locations , through continuous learning from real-time traffic conditions, station utilization and grid constraints. This would result in more efficient load balancing, reduced waiting times and decreased huge-pressue on the electrical grid. Ultimately this will be contributing to a more resilient and intelligent charging infrastructures.

Expansion of Simulation Environment: To further enhance the generalizability and realism of this research work, future research should consider the extension of the simulation environment to represent larger and more complex urban systems. This can include- additional road networks, junctions and traffic control systems. As well, modeling a wider variety of vehicle types with diverse behavioral characteristics can be included in an improved version of this work. Including randomfactors such as : suddentraffic jams, accidents or sudden demand increases will make the simulation more representative of real-world situations. Such improvements will allow researchers to evaluate the performance of load forecasting models under diverse and uncertain conditions. This will increase their robustness and applicability in different urban contexts.

Integration of Renewable Energy: Renewable energy sources integration into the Electric Vehicle charging ecosystem can be an important expansion of this study. Future research could model charging stations powered by solar panels or hybrid power systems . Also could model analyzing how renew able energy generation can be effectively used to meet charging demand. It would require researching the temporal balance between energy generation and consumption, as well as developing energy storage and load shifting methods. Into the simulation framework , by bringing renewable energy will push transportation toward sustainability . And it will reduce dependence on the traditional power grid, making the whole setup more environmentally friendly and cost effective.

Real-Time System Deployment: In the last future research should focus on moving from simulation based testing to deploying real time systems. This requires integrating advanced machine learning models into Internet of Things (IoT -enabled) smart grid infrastructures. Where data can be continuously collected, processed and analyzed. Utilizing robust integrated frameworks like DeepBoost [3] can offer the necessary precision for such real-time monitors. Putting real time deployment into action means keeping an eye on charging demand as it happens, predicting future load patterns and using automated controls to manage resources more efficiently. Incorporating these systems will allow service organizations and policymakers to make more informed decisions. Moreover it'll allow directly connecting academic research to the realities of transportation and energy management.

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